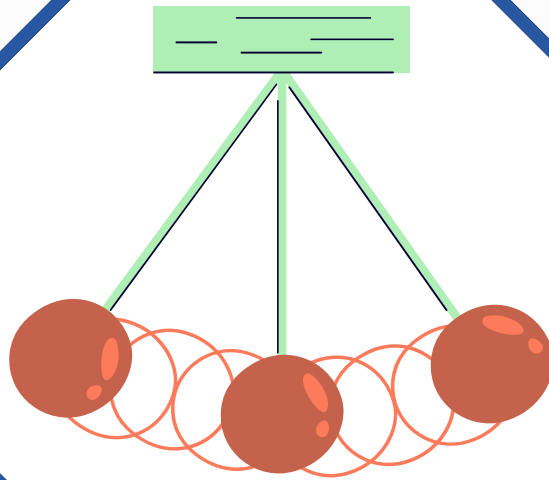


JEE (MAIN+ADVANCED)

Physics

NURTURE COURSE



STUDY MATERIAL

Simple Harmonic Motion

English Medium

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SIMPLE HARMONIC MOTION

1.0 PERIODIC MOTION

SL AL

Any motion which repeats itself after regular interval of time is called periodic motion or harmonic motion. The constant interval of time after which the motion is repeated is called time period.

Examples : (i) Motion of planets around the sun. (ii) Motion of the pendulum of wall clock.

2.0 OSCILLATORY MOTION

SL AL

The motion of body is said to be oscillatory or vibratory motion if it moves back and forth (to and fro) about a fixed point after regular interval of time.

The fixed point about which the body oscillates is called mean position or equilibrium position.

Examples :

- (i) Vibration of the wire of 'Sitar'.
- (ii) Oscillation of the mass suspended from spring.

3.0 SIMPLE HARMONIC MOTION (S.H.M.)

SL AL

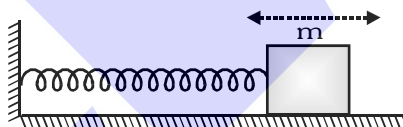
Simple harmonic motion is the simplest form of vibratory or oscillatory motion.

3.1 S.H.M. are of two types

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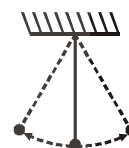
Linear S.H.M.

When a particle moves to and fro about a fixed point (called equilibrium position) along a straight line then its motion is called linear simple harmonic motion.



Angular S.H.M.

When a system oscillates angularly with respect to a fixed axis then its motion is called angular simple harmonic motion. Illustration :- Motion of a bob of simple pendulum.



3.2. Necessary Condition to execute S.H.M.

SL AL

In Linear S.H.M.

The resultant force (or acceleration) acting on the particle should always be proportional to the displacement of the particle and directed towards the equilibrium position

$$\therefore \vec{F} \propto -\vec{x} \text{ or } \vec{a} \propto -\vec{x}$$

Negative sign shows that direction of force and acceleration is towards equilibrium position and x is displacement of particle from equilibrium position.

In Angular S.H.M.

The resultant torque (or angular acceleration) acting on the particle should always be proportional to the angular displacement of the particle and directed towards the equilibrium position

$$\therefore \vec{\tau} \propto -\vec{\theta} \text{ or } \vec{\alpha} \propto -\vec{\theta}$$

3.3. Equation of simple Harmonic motion

SL AL

In linear S.H.M.

Restoring force acting on the particle,

$$\vec{F} = ma = -ka \Rightarrow a = -\frac{kx}{m} \Rightarrow \frac{dv}{dt} = -\frac{kx}{m} \Rightarrow v \frac{dv}{dt} = -\frac{kx}{m} \frac{dx}{dt} \quad [\because v = dx/dt]$$

$$\int v \, dv = -\int \frac{kx}{m} \, dx \Rightarrow \frac{v^2}{2} = -\frac{kx^2}{2m} + C$$

$$\text{At } x = 0, v = v_0 \therefore C = \frac{v_0^2}{2} \Rightarrow \frac{v^2}{2} = -\frac{\omega^2 x^2}{2} + \frac{v_0^2}{2} \quad [\because \omega^2 = k/m]$$

$$\Rightarrow v = \sqrt{v_0^2 - \omega^2 x^2} \Rightarrow \frac{dx}{dt} = \sqrt{v_0^2 - \omega^2 x^2}$$

$$\Rightarrow \int \frac{dx}{\sqrt{v_0^2 - \omega^2 x^2}} = \int dt \Rightarrow \frac{1}{\omega} \sin^{-1} \left(\frac{\omega x}{v_0} \right) = t + C_1 \quad [\because v_0 = \omega A] \Rightarrow x = A \sin(\omega t + \omega C_1)$$

$\therefore$ At $t = 0, x = 0$ and if velocity is in $+x$ direction $x = A \sin \omega t$

if velocity is in $-x$ direction, $x = -A \sin \omega t$

In Angular SHM

Restoring torque acting on the $\vec{\tau} = -C\vec{\theta}$ particle where C is a constant which can be defined as torque per unit angular displacement.

Mathematically, $I\ddot{\alpha} = -C\vec{\theta}$. Where I is the moment of inertia of the system about the axis of rotation.

$$\Rightarrow I \frac{d^2\theta}{dt^2} + C\theta = 0 \Rightarrow \frac{d^2\theta}{dt^2} + \left(\frac{C}{I}\right)\theta = 0. \text{ Since, } \frac{d^2\theta}{dt^2} + \omega^2\theta = 0 \Rightarrow \omega = \sqrt{\left(\frac{C}{I}\right)} \quad \dots(i)$$

Illustrations

Illustration 1. If equation of displacement of a particle is $y = A \sin \omega t + B \cos \omega t$ then find the nature of the motion of particle.

Solution $y = A \sin \omega t + B \cos \omega t$

Differentiate with respect to t

$$\frac{dy}{dt} = A\omega \cos \omega t - B\omega \sin \omega t$$

Again differentiating with respect to t

$$\frac{d^2y}{dt^2} = -\omega^2 A \sin \omega t - \omega^2 B \cos \omega t$$

$$\frac{d^2y}{dt^2} = -\omega^2 (A \sin \omega t + B \cos \omega t)$$

$$\Rightarrow \frac{d^2y}{dt^2} = -\omega^2 y \Rightarrow \frac{d^2y}{dt^2} + \omega^2 y = 0$$

It is a differential equation of linear S.H.M. So motion of the particle is simple harmonic

3.4 Some basic terms

SL AL

Mean Position

The point at which the restoring force / torque on the particle is zero and potential energy is minimum, is known as its mean position.

Restoring Force (F)

- The force acting on the particle which tends to bring the particle towards its mean position, is known as restoring force.

- Restoring force always acts in a direction opposite to that of displacement. Displacement is measured from the mean position.

Amplitude (A)

The maximum (positive or negative) value of displacement of particle from mean position is define as amplitude. Since maximum and minimum value of sin or cos function is +1& -1, therefore here amplitude is A.

Time period (T)

- The minimum time after which the particle keeps on repeating its motion is known as time period.
- The smallest time taken to complete one oscillation or vibration is also define as time period.
- It is given by $T = \frac{2\pi}{\omega} = \frac{1}{n}$ where ω is angular frequency and n is frequency.

Oscillation or Vibration

When a particle goes on one side from mean position and returns back and then it goes to other side and again returns back to mean position, then this process is known as one oscillation.

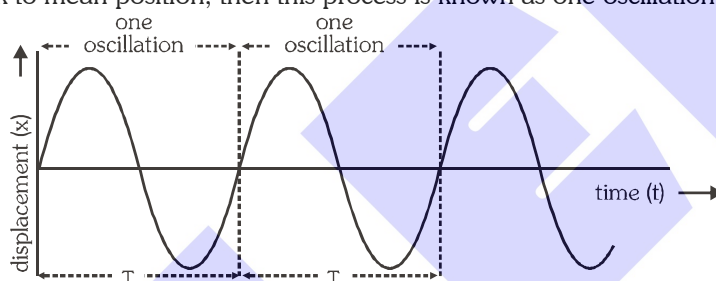


Figure - 3 :

Frequency (n or f)

- The number of oscillations per second is define as frequency.
- It is given by $n = \frac{1}{T}$, $n = \frac{\omega}{2\pi}$
- SI UNIT** : hertz (Hz), 1 hertz = 1 cycle per second (cycle is a number not a dimensional quantity).
- Dimensions** : $M^0 L^0 T^{-1}$.

Phase

- Phase of a vibrating particle at any instant is the state of the vibrating particle regarding its displacement and direction of vibration at that particular instant.
- In the equation $x = A \sin (\omega t + \theta)$, $(\omega t + \theta)$ is the phase of the particle.
- The phase angle at time $t = 0$ is known as initial phase or epoch.
- The difference of total phase angles of two particles executing S.H.M. with respect to the mean position is known as phase difference.
- Two vibrating particles are said to be in same phase if the phase difference between them is an even multiple of π , i.e., $\Delta\phi = 2n\pi$ Where $n = 0, 1, 2, 3, \dots$
- Two vibrating particle are said to be in opposite phase if the phase difference between them is an odd multiple of π i.e., $\Delta\phi = (2n + 1)\pi$ Where $n = 0, 1, 2, 3, \dots$

Angular frequency (ω)

The rate of change of phase angle of a particle with respect to time is define as its angular frequency.

SI unit : radian/second, **Dimensions** : $M^0 L^0 T^{-1}$, $\omega = \sqrt{\frac{k}{m}}$

Illustrations

Illustration 2. If two S.H.M.'s are represented by equations $y_1 = 10 \sin \left[3\pi t + \frac{\pi}{4} \right]$ and

$$y_2 = 5 \left[\sin(3\pi t) + \sqrt{3} \cos(3\pi t) \right]$$

then find the ratio of their amplitudes and phase difference in between them.

Solution As $y_2 = 5 \left[\sin(3\pi t) + \sqrt{3} \cos(3\pi t) \right]$... (i)

So if $5 = A \cos \phi$ and

$$\text{Then } A = \sqrt{5^2 + (5\sqrt{3})^2} = 10 \text{ and } \tan \phi = \frac{5\sqrt{3}}{5} = \sqrt{3} \text{ so } \phi = \frac{\pi}{3}$$

The above equation (i) becomes $y_2 = A \cos \phi \sin(3\pi t) + A \sin \phi \cos(3\pi t) \Rightarrow y_2 = A \sin(3\pi t + \phi) = 10$

$$\sin \left[3\pi t + \left(\frac{\pi}{3} \right) \right] \text{ so, } \frac{A_1}{A_2} = \frac{10}{10} \Rightarrow A_1 : A_2 = 1 : 1, \text{ Phase difference} = \frac{\pi}{4} - \frac{\pi}{3} = -\frac{\pi}{12}$$

3.5. Displacement in S.H.M.

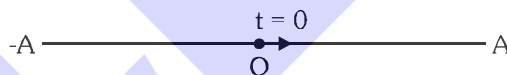
SL AL

- The displacement of a particle executing linear S.H.M. at any instant is defined as the distance of the particle from the mean position at that instant.
- It can be given by relation $x = A \sin \omega t$ or $x = A \cos \omega t$.

The first relation is valid when the time is measured from the mean position and the second relation is valid when the time is measured from the extreme position of the particle executing S.H.M. along a straight line path.

Illustrations

Illustration 3. A particle starts from mean position and moves towards positive extreme as shown below. Find the equation of the SHM. Amplitude of SHM is A.

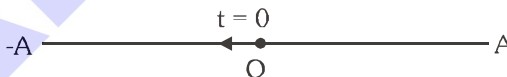


Solution General equation of SHM can be written as $x = A \sin(\omega t + \phi)$

$$\text{At } t=0, x=0 \quad \therefore 0 = A \sin \phi \quad \therefore \phi = 0, \pi \quad \phi \in [0, 2\pi]$$

$$\text{Also, at } t=0, v = +ve \quad \therefore A\omega \cos \phi = \text{maximum } +ve \text{ or } \phi = 0$$

Hence, if the particle is at mean position at $t=0$ and is moving towards +ve extreme, then the equation of SHM is given by $x = A \sin \omega t$



Similarly, for particle moving towards -ve extreme then $\phi = \pi$

$\therefore$ equation of SHM is $x = A \sin(\omega t + \pi)$ or, $x = -A \sin \omega t$

Illustration 4. Write the equation of SHM for the situation show below :



Solution General equation of SHM can be written as $x = A \sin(\omega t + \phi)$

$$\text{At } t=0, x=A/2 \Rightarrow \frac{A}{2} = A \sin \phi \Rightarrow \phi = 30^\circ, 150^\circ$$

$$\text{Also at } t=0, v = -ve; A\omega \cos \phi = -ve \Rightarrow \phi = 150^\circ$$

3.6. Velocity in SHM

SL AL

- (i) It is define as the time rate of change of the displacement of the particle at the given instant.

- (ii) Velocity in S.H.M. is given by $v = \frac{dx}{dt} = \frac{d}{dt}(A \sin \omega t) \Rightarrow v = A\omega \cos \omega t$

$$v = \pm A\omega \sqrt{1 - \sin^2 \omega t} \Rightarrow v = \pm A\omega \sqrt{1 - \frac{x^2}{A^2}} = \pm \omega \sqrt{A^2 - x^2} \quad [\because x = A \sin \omega t]$$

Squaring both the sides $v^2 = \omega^2 (A^2 - x^2)$

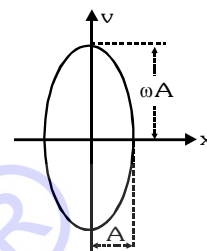
$$\Rightarrow \frac{v^2}{\omega^2} = A^2 - x^2 \Rightarrow \frac{v^2}{\omega^2 A^2} = 1 - \frac{x^2}{A^2} \Rightarrow \frac{x^2}{A^2} + \frac{v^2}{A^2 \omega^2} = 1$$

This is equation of ellipse. So curve between displacement and velocity of particle executing S.H.M. is ellipse.

- (iii) The graph between velocity and displacement is shown in figure. If particle oscillates with unit angular frequency ($\omega = 1$) then curve between v and x will be circular.

Note

- (i) The direction of velocity of a particle in S.H.M. is either towards or away from the mean position.
 (ii) At mean position ($x = 0$), speed is maximum ($= A\omega$) and at extreme position ($x = \pm A$), the velocity of particle executing S.H.M. is zero (minimum).



3.7. Acceleration in SHM

SL AL

- (i) It is define as the time rate of change of the velocity of the particle at given instant.

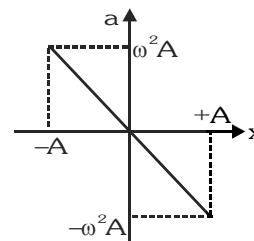
- (ii) Acceleration in S.H.M. is given by $a = \frac{dv}{dt} = \frac{d}{dt}(A\omega \cos \omega t)$

$$a = -\omega^2 A \sin \omega t \Rightarrow a = -\omega^2 x$$

- (iii) The graph between acceleration and displacement as shown in figure

Note

- (i) The acceleration of a particle executing S.H.M. is always directed towards the mean position.
 (ii) The magnitude of acceleration of the particle executing S.H.M. is maximum at extreme position ($= \omega^2 A$) and minimum at mean position ($= \text{zero}$)



4.0 GEOMETRICAL MEANING OF S.H.M.

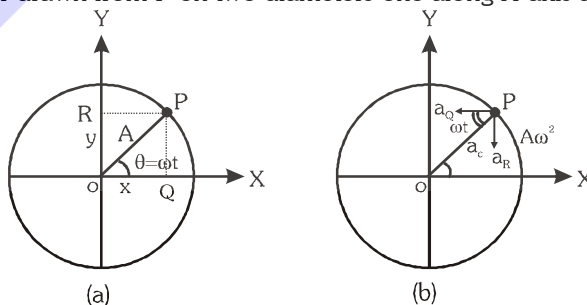
SL AL

If a particle is moving with uniform speed along the circumference of a circle then the straight line motion of the foot of perpendicular drawn from the particle on the diameter of the circle is called 'S.H.M.'

4.1. SHM as a Projection of Circular Motion

SL AL

Suppose a particle P is moving uniformly on a circle of radius A with angular speed ω . Q and R be the two feet of the perpendicular drawn from P on two diameters one along X-axis and the other along Y-axis.



Suppose the particle P is on the X-axis at $t=0$. Radius OP makes an angle $\theta = \omega t$ with the X-axis at time t then
 $x = A \cos \omega t$, $y = A \sin \omega t$

Here, x and y are the displacement of Q and R from the origin at time t , which are the displacement equations of SHM. It implies that although P is under uniform circular motion, Q and R are performing SHM about O with the same angular speed ω as that of P . From figure (b) centripetal acceleration of $P = a_c = A\omega^2$ (towards the centre) a_c can have two components as shown in the figure $a_R = A\omega^2 \sin\omega t = \omega^2 x$, $a_Q = A\omega^2 \cos\omega t = \omega^2 y$ and a_R and a_Q are actually the acceleration corresponding to the points R and Q respectively.

4.2. Comparison between linear and angular S.H.M.

Linear S.H.M.	Angular S.H.M.
$\vec{F} \propto \vec{x} \Rightarrow \vec{F} = -k\vec{x}$ where k is the restoring force constant $\vec{a} = -\frac{k}{m}\vec{x} \Rightarrow \frac{d^2\vec{x}}{dt^2} + \frac{k}{m}\vec{x} = 0$ It is known as differential equation of linear SHM. $x = A \sin\omega t$; $a = -\omega^2 x$ where ω is the angular frequency $\omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}} = \frac{2\pi}{T} = 2\pi n$ where T is time period and n is frequency $T = 2\pi\sqrt{\frac{m}{k}}$, $n = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$ This concept is valid for all types of linear S.H.M.	$\vec{\tau} \propto -\vec{\theta} \Rightarrow \vec{\tau} = -C\vec{\theta}$ where C is the restoring torque constant $\vec{\alpha} = -\frac{C}{I}\vec{\theta} \Rightarrow \frac{d^2\vec{\theta}}{dt^2} + \frac{C}{I}\vec{\theta} = 0$ It is known as differential equation of angular SHM. $\theta = \theta_0 \sin\omega t$; $\alpha = -\omega^2\theta$ $\omega^2 = \frac{C}{I} \Rightarrow \omega = \sqrt{\frac{C}{I}} = \frac{2\pi}{T} = 2\pi n$ where T is time period and n is frequency $T = 2\pi\sqrt{\frac{I}{C}}$, $n = \frac{1}{2\pi}\sqrt{\frac{C}{I}}$ This concept is valid for all types of angular SHM.

4.3. Graphical representation

SL AL

Graphical study of displacement, velocity, acceleration and force in S.H.M.

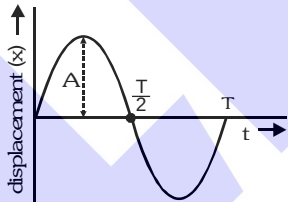
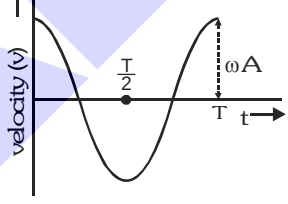
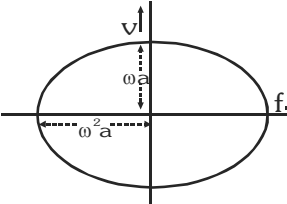
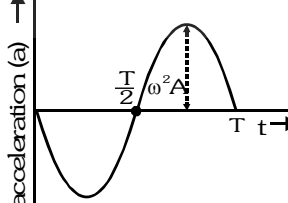
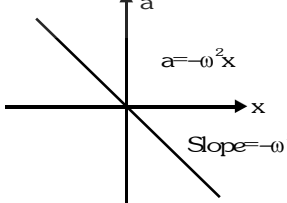
S. No.	Graph	In form of t	In form of x	Maximum value
	Displacement	$x = A \sin\omega t$ 	$x = x$	$x = \pm A$
	Velocity	$v = A\omega \cos\omega t$ 	$v = \pm\omega\sqrt{A^2 - x^2}$ 	$v = \pm\omega A$
	Acceleration	$a = -\omega^2 A \sin\omega t$ 	$a = -\omega^2 x$  (a) Acceleration v/s displacement	$a = \pm \omega^2 A$
	Force ($F = ma$)	$F = -m\omega^2 A \sin\omega t$	$F = -m\omega^2 x$	$F = \pm m\omega^2 A$

Illustration 8. Periodic time of a simple pendulum is 2 second and it can travel to and fro from equilibrium position upto maximum 5cm. At start the pendulum is at maximum displacement on right side of equilibrium position. Find displacement and time relation.

Solution Displacement expression for S.H.M. , $x = A \sin (\omega t + \phi)$

$$\text{Time period of simple pendulum } T = \frac{2\pi}{\omega} = 2 \text{ s} \quad \therefore \omega = \pi \text{ rad/s}$$

$$\text{Amplitude of pendulum } A = 5 \text{ cm} \quad \therefore x = 5 \sin (\pi t + \phi)$$

$$\text{Now at } t = 0, \text{ displacement } x = 5 \text{ cm} \quad \therefore 5 = 5 \sin (\pi \times 0 + \phi) \Rightarrow \sin \phi = 1 \Rightarrow \phi = \frac{\pi}{2}$$

$$\text{Therefore, } x = 5 \sin \pi \left(t + \frac{1}{2}\right) \Rightarrow x = 5 \sin \left(\pi t + \frac{\pi}{2}\right) \Rightarrow x = 5 \cos \pi t$$

Illustration 9. The velocity of a particle in S.H.M. at positions x_1 and x_2 are v_1 and v_2 respectively. Determine value of time period and amplitude.

Solution $v = \omega \sqrt{A^2 - x^2} \Rightarrow v^2 = \omega^2 (A^2 - x^2)$

$$\text{At position } x_1, v_1^2 = \omega^2 (A^2 - x_1^2) \quad \dots(i)$$

$$\text{At position } x_2, v_2^2 = \omega^2 (A^2 - x_2^2) \quad \dots(ii)$$

$$\text{Subtracting (ii) from (i)} \quad v_1^2 - v_2^2 = \omega^2 (x_2^2 - x_1^2) \Rightarrow \omega = \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}}$$

$$\text{Time period } T = \frac{2\pi}{\omega} \Rightarrow T = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$$

$$\text{Dividing (i) by (ii)} \quad \frac{v_1^2}{v_2^2} = \frac{A^2 - x_1^2}{A^2 - x_2^2} \Rightarrow v_1^2 A^2 - v_1^2 x_2^2 = v_2^2 A^2 - v_2^2 x_1^2$$

$$\text{So } A^2 (v_1^2 - v_2^2) = v_1^2 x_2^2 - v_2^2 x_1^2 \Rightarrow A = \sqrt{\frac{v_1^2 x_2^2 - v_2^2 x_1^2}{v_1^2 - v_2^2}}$$

Illustration 10. A particle executing S.H.M. having amplitude 0.01 m and frequency 60 Hz. Determine maximum acceleration of particle.

Solution Maximum acceleration $a_{\max} = \omega^2 A = 4\pi^2 n^2 A = 4\pi^2 (60)^2 \times (0.01) = 144 \pi^2 \text{ m/s}^2$

BEGINNER'S BOX-1

General Equation of SHM

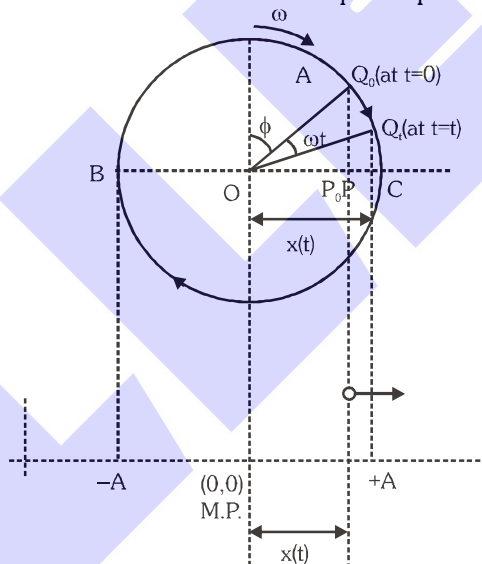
- A particle is oscillating according to the equation $X = 7 \cos 0.5\pi t$, where t is in second. The point moves from the position of equilibrium to maximum displacement in time
(A) 4.0 sec (B) 2.0 sec (C) 1.0 sec (D) 0.5 sec
- A simple harmonic oscillator has an amplitude a and time period T . The time required by it to travel from $x = a$ to $x = a/2$ is
(A) $T/6$ (B) $T/4$ (C) $T/3$ (D) $T/2$
- A particle of mass 10 grams is executing simple harmonic motion with an amplitude of 0.5 m and periodic time of $(\pi/5)$ seconds. The maximum value of the force acting on the particle is
(A) 25 N (B) 5 N (C) 2.5 N (D) 0.5 N
- A body is executing S.H.M. When its displacement from the mean position is 4 cm and 5 cm, the corresponding velocity of the body is 10 cm/sec and 8 cm/sec. Then the time period of the body is
(A) 2π sec (B) $\pi/2$ sec (C) π sec (D) $3\pi/2$ sec

5. A particle executes simple harmonic motion with an amplitude of 4 cm. At the mean position the velocity of the particle is 10 cm/s. The distance of the particle from the mean position when its speed becomes 5 cm/s is
 (A) $\sqrt{3}$ cm (B) $\sqrt{5}$ cm (C) $2(\sqrt{3})$ cm (D) $2(\sqrt{5})$ cm
6. A particle executes linear simple harmonic motion with an amplitude of 2 cm. When the particle is at 1 cm from the mean position the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is
 (A) $\frac{1}{2\pi\sqrt{3}}$ (B) $2\pi\sqrt{3}$ (C) $\frac{2\pi}{\sqrt{3}}$ (D) $\frac{\sqrt{3}}{2\pi}$
7. In simple harmonic motion, the ratio of acceleration of the particle to its displacement at any time is a measure of
 (A) Spring constant (B) Angular frequency (C) (Angular frequency)² (D) Restoring force

5.0 SHM AS A PROJECTION OF UNIFORM CIRCULAR MOTION

SL AL

Consider a particle Q, moving on a circle of radius A with constant angular velocity ω . The projection of Q on a diameter BC is P. It is clear from the figure that as Q moves around the circle the projection P executes a simple harmonic motion on the x-axis between B and C. The angle that the radius OQ makes with the +ve vertical in clockwise direction in at $t = 0$ is equal to phase constant (ϕ).



Let the radius OQ_0 makes an angle ωt with the OQ_t at time t . Then $x(t) = A \sin(\omega t + \phi)$

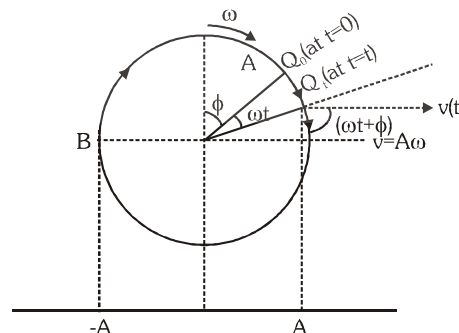
In the above discussion the foot of projection is x-axis so it is called horizontal phasor. Similarly the foot of perpendicular on y-axis will also executes SHM of amplitude A and angular frequency ω [$y(t) = A \cos \omega t$]. This is called vertical phasor. The phasor of the two SHM differ by $\pi/2$.

5.1. Problem solving strategy in horizontal phasor

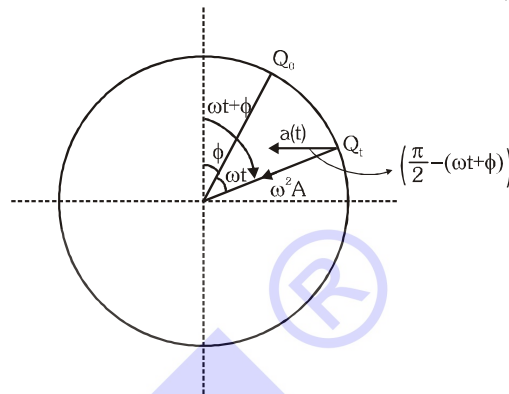
SL AL

1. First assume circle of radius equal to amplitude of SHM.
2. Assume a particle rotating in a circular path moving with constant ω same as that of SHM in clockwise direction.
3. Angle made by the particle at $t=0$ with the upper vertical is equal to phase constant.
4. Horizontal component of velocity of particle gives you the velocity of particle performing SHM for example from figure

$$v(t) = A\omega \cos(\omega t + \phi)$$

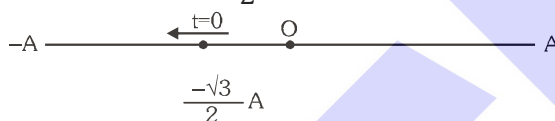


5. Component of acceleration of particle in horizontal direction is equals to the acceleration of particle performing SHM. The acceleration of a particle in uniform circular motion is only centripetal and has a magnitude $a = \omega^2 A$ from figure $a(t) = -\omega^2 A \sin(\omega t + \phi)$



Illustrations

Illustration 11*. A particle starts from point $x = -\frac{\sqrt{3}}{2} A$ and move towards negative extreme as shown following



- (a) Find the equation of the SHM
 (b) Find the time taken by the particle to go directly from its initial position to negative extreme.
 (c) Find the time taken by the particle to reach at mean position.
 (a) Figure shows the solution of the problem with the help of phasor. Horizontal component of velocity at Q gives the required direction of velocity at $t = 0$ so we will choose it.

$$\text{In } \triangle OSQ : \cos \theta = \frac{\sqrt{3}/2A}{A} = \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}$$

$$\text{Now } \phi = \frac{3\pi}{2} - \frac{\pi}{6} = \frac{8\pi}{6} = \frac{4\pi}{3}$$

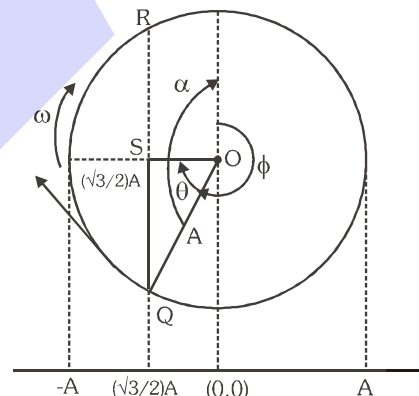
$$\text{So equation of SHM is } x = A \sin\left(\omega t + \frac{4\pi}{3}\right)$$

- (b) Now to reach the particle at left extreme point it will travel angle θ along the circle. So time taken.

$$t = \frac{\pi}{\omega} = \frac{\pi}{6\omega} \Rightarrow t = \frac{T}{12} \text{ sec}$$

- (c) To reach the particle at mean position it will travel angle $\alpha = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3}$

$$\text{So, time taken} = \frac{\alpha}{\omega} = \frac{T}{3}$$



6.0 ENERGY OF PARTICLE IN S.H.M.

6.1. Potential Energy (U or P.E.)

SL AL

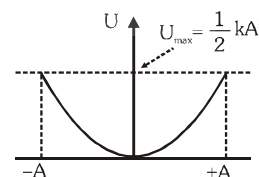
(i) In terms of displacement

The potential energy is related to force by the relation $F = -\frac{dU}{dx} \Rightarrow \int dU = -\int F dx$

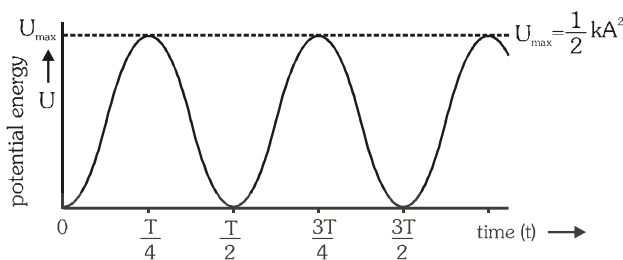
For S.H.M. $F = -kx$ so $\int dU = -\int (-kx) dx = \int kx dx \rightarrow U = \frac{1}{2} kx^2 + C$

At $x = 0$, $U = U_0 \Rightarrow C = U_0$ So $U = \frac{1}{2} kx^2 + U_0$

Where the potential energy at equilibrium position = U_0 when $U_0 = 0$ then $U = 1/2 kx^2$



(ii) In terms of time



Since $x = A \sin(\omega t + \phi)$, $U = \frac{1}{2} kA^2 \sin^2(\omega t + \phi)$

If initial phase (ϕ) is zero then $U = \frac{1}{2} kA^2 \sin^2 \omega t = \frac{1}{2} m\omega^2 A^2 \sin^2 \omega t$

Note

- (i) In S.H.M. the potential energy is a parabolic function of displacement, the potential energy is minimum at the mean position ($x = 0$) and maximum at extreme position ($x = \pm A$)
- (ii) The potential energy is the periodic function of time.

It is minimum at $t = 0, \frac{T}{2}, T, \frac{3T}{2}, \dots$ and maximum at $t = \frac{T}{4}, \frac{3T}{4}, \frac{5T}{4}, \dots$

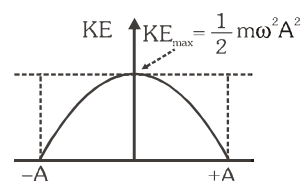
6.2. Kinetic Energy (K)

SL AL

(i) In terms of displacement

If mass of the particle executing S.H.M. is m and its velocity is v then kinetic energy at any instant.

$$K = \frac{1}{2} mv^2 = \frac{1}{2} m\omega^2 (A^2 - x^2) = \frac{1}{2} k(A^2 - x^2)$$



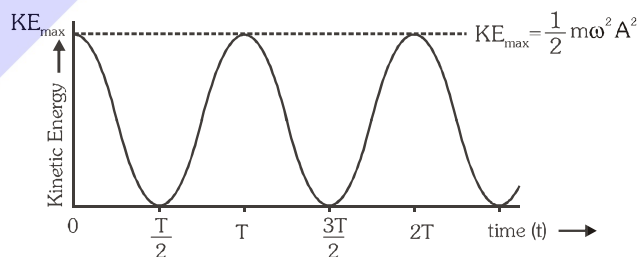
(ii) In terms of time

$$v = A\omega \cos(\omega t + \phi)$$

$$K = \frac{1}{2} m\omega^2 A^2 \cos^2(\omega t + \phi)$$

If initial phase ϕ is zero

$$K = \frac{1}{2} m\omega^2 A^2 \cos^2 \omega t$$



Note

- (i) In S.H.M. the kinetic energy is an inverted parabolic function of displacement. The kinetic energy is maximum ($\frac{1}{2} kA^2$) at mean position ($x = 0$) and minimum (zero) at extreme position ($x = \pm A$)
- (ii) The kinetic energy is the periodic function of time. It is maximum at $t = 0, T, 2T, 3T, \dots$ and minimum at $t = \frac{T}{2}, \frac{3T}{2}, \frac{5T}{2}, \dots$

6.3. Total energy (E)

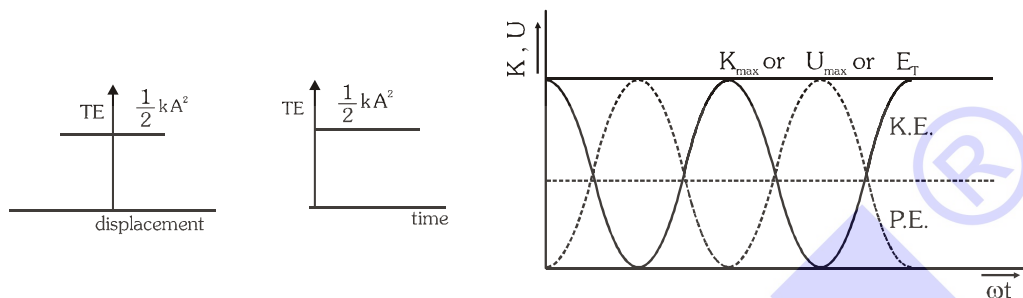
SL AL

Total energy in S.H.M. is given by ; $E = \text{potential energy} + \text{kinetic energy} = U + K$

- (i) w.r.t. position $E = \frac{1}{2} kx^2 + \frac{1}{2} k(A^2 - x^2) \Rightarrow E = \frac{1}{2} kA^2 = \text{constant}$

(ii) w.r.t. time

$$E = \frac{1}{2} m\omega^2 A^2 \sin^2 \omega t + \frac{1}{2} m\omega^2 A^2 \cos^2 \omega t = \frac{1}{2} m\omega^2 A^2 (\sin^2 \omega t + \cos^2 \omega t) = \frac{1}{2} m\omega^2 A^2 = \frac{1}{2} kA^2 \text{ constant}$$

**Note**

- (i) Total energy of a particle in S.H.M. is same at all instant and at all displacement.
- (ii) Total energy depends upon mass, amplitude and frequency of vibration of the particle executing S.H.M.

6.4. Average energy in S.H.M.**SL AL**

(i) The time average of P.E. and K.E. over one cycle is

$$(a) \langle KE \rangle_t = \langle \frac{1}{2} m\omega^2 A^2 \cos^2 \omega t \rangle = \frac{1}{2} m\omega^2 A^2 \langle \cos^2 \omega t \rangle = \frac{1}{2} m\omega^2 A^2 \left(\frac{1}{2} \right) = \frac{1}{4} m\omega^2 A^2 = \frac{1}{4} kA^2$$

$$(b) \langle PE \rangle_t = \langle \frac{1}{2} m\omega^2 A^2 \sin^2 \omega t \rangle = \frac{1}{2} m\omega^2 A^2 \langle \sin^2 \omega t \rangle = \frac{1}{2} m\omega^2 A^2 \left(\frac{1}{2} \right) = \frac{1}{4} m\omega^2 A^2 = \frac{1}{4} kA^2$$

$$(c) \langle TE \rangle_t = \langle \frac{1}{2} m\omega^2 A^2 + U_0 \rangle = \frac{1}{2} m\omega^2 A^2 + U_0 = \frac{1}{2} kA^2 + U_0$$

(ii) The position average of P.E. and K.E. between $x = -A$ to $x = A$

$$(a) \langle KE \rangle_x = \frac{\int_{-A}^A \frac{1}{2} m\omega^2 (A^2 - x^2) dx}{\int_{-A}^A dx} = \frac{\frac{1}{2} m\omega^2 \left(A^2 x - \frac{x^3}{3} \right)_{-A}^A}{2A} = \frac{1}{3} kA^2$$

$$(b) \langle PE \rangle_x = \frac{\int_{-A}^A (PE) dx}{\int_{-A}^A dx} = \frac{\int_{-A}^A \left(U_0 + \frac{1}{2} kx^2 \right) dx}{\int_{-A}^A dx} = \frac{U_0 (2A) + \frac{1}{2} k \left(\frac{x^3}{3} \right)_{-A}^A}{2A} = U_0 + \frac{1}{6} kA^2$$

$$(c) \langle TE \rangle_x = \frac{\int_{-A}^A (TE) dx}{\int_{-A}^A dx} = \frac{\int_{-A}^A \left(\frac{1}{2} kA^2 + U_0 \right) dx}{\int_{-A}^A dx} = \frac{1}{2} kA^2 + U_0$$

Illustrations

Illustration 12. In case of simple harmonic motion

- What fraction of total energy is kinetic and what fraction is potential when displacement is one half of the amplitude.
- At what displacement the kinetic and potential energies are equal.

Solution In S.H.M. : Kinetic Energy $K = \frac{1}{2}k(A^2 - x^2)$, Potential Energy $U = \frac{1}{2}kx^2$,
Total Energy (TE) = $\frac{1}{2}kA^2$

$$(a) \text{ Fraction of Kinetic Energy } f_{K.E.} = \frac{K}{T.E.} = \frac{A^2 - x^2}{A^2},$$

$$\text{Fraction of Potential Energy } f_{P.E.} = \frac{U}{T.E.} = \frac{x^2}{A^2}$$

$$\text{at } x = \frac{A}{2} \quad f_K = \frac{A^2 - A^2/4}{A^2} = \frac{3}{4} \quad \text{and } f_U = \frac{A^2/4}{A^2} = \frac{1}{4}$$

$$(b) K = U \Rightarrow \frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2 \Rightarrow 2x^2 = A^2 \Rightarrow x = \pm \frac{A}{\sqrt{2}}$$

Illustration 13. The potential energy of a particle oscillating on x-axis is $U = 20 + (x - 2)^2$. Here U is in joules and x in meters. Total mechanical energy of the particle is 36 J.

- State whether the motion of the particle is simple harmonic or not.
- Find the mean position.
- Find the maximum kinetic energy of the particle.

Solution (a) $F = -\frac{dU}{dx} = -2(x - 2)$ By assuming $x - 2 = X$, we have $F = -2X$

Since, $F \propto -X$. The motion of the particle is simple harmonic

- The mean position of the particle is $X = 0 \Rightarrow x - 2 = 0$, which gives $x = 2\text{m}$
- Maximum kinetic energy of the particle is, $K_{\max} = E - U_{\min} = 36 - 20 = 16\text{ J}$

Note : $U_{\min}$ is 20 J at mean position or at $x = 2\text{m}$.

BEGINNER'S BOX-2

Energy of Particle in SHM

- A student says that he had applied a force $F = -k\sqrt{x}$ on a particle and the particle moved in simple harmonic motion. He refuses to tell whether k is a constant or not. Assume that he has worked only with positive x and no other force acted on the particle.
 - As x increases k increases
 - As x increases k decreases
 - As x increases k remains constant.
 - The motion cannot be simple harmonic.
- The displacement (in m) of a particle of mass 100 g from its equilibrium position is given by the equation:

$$y = 0.05 \sin 3\pi (5t + 0.4)$$
 - the time period of motion is 1/30 sec
 - the time period of motion is 1/7.5 sec
 - the maximum acceleration of the particle is $11.25\pi^2 \text{ m/s}^2$
 - the force acting on the particle is zero when the displacement is 0.05 m.
- The magnitude of average acceleration in half time period from equilibrium position in a simple harmonic motion is
 - $\frac{2A\omega^2}{\pi}$
 - $\frac{A\omega^2}{2\pi}$
 - $\frac{A\omega^2}{\sqrt{2}\pi}$
 - Zero

4. In a simple harmonic motion
 (A) the potential energy is always equal to the kinetic energy
 (B) the potential energy is never equal to the kinetic energy
 (C) the average potential energy in any time interval is equal to the average kinetic energy in that time interval
 (D) the average potential energy in one time period is equal to the average kinetic energy in this period.
5. A particle executes simple harmonic motion with a frequency ν . The frequency with which the kinetic energy oscillates is
 (A) $\nu/2$ (B) ν (C) 2ν (D) zero
6. A simple harmonic motion has an amplitude A and time period T . Find the time required by it to travel directly from
 (A) $x = 0$ to $x = A/2$ (B) $x = 0$ to $x = \frac{A}{\sqrt{2}}$ (C) $x = A$ to $x = A/2$ (D) $x = -\frac{A}{\sqrt{2}}$ to $x = \frac{A}{\sqrt{2}}$
 (E) $x = \frac{A}{\sqrt{2}}$ to $x = A$.
7. A particle executes simple harmonic motion with an amplitude of 10 cm and time period 6 s. At $t = 0$ it is at position $x = 5$ cm going towards positive x -direction. Write the equation for the displacement x at time t . Find the magnitude of the acceleration of the particle at $t = 4$ s.
8. Calculate the ratio between the kinetic energy E_k and the potential energy E_p of an oscillating point for the moment $t = T/12$, where T = time period.

7.0 SPRING SYSTEM

SL AL



helical spring

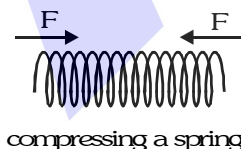
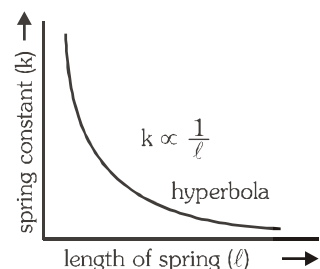
spiral spring



- (i) When spring is given small displacement by stretching or compressing it, then restoring elastic force is developed in it because it obeys Hook's law.

$$\vec{F} \propto -\vec{x} \Rightarrow \vec{F} = -k\vec{x} \quad \text{Here } k \text{ is spring constant}$$

- (ii) Spring is assumed massless, so restoring elastic force in spring is assumed same everywhere.
- (iii) Spring constant (k) depends on length (ℓ), radius and material of wire used in spring. for spring $k\ell = \text{constant}$
- (iv) When spring is compressed or stretched then work done on it is stored as elastic potential energy.



compressing a spring



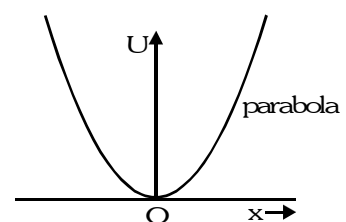
spring without deformation



stretching a spring

$$W = \int F dx = \int kx dx \quad \text{and} \quad U = W = \frac{1}{2} kx^2$$

When spring is stretched from ℓ_1 to ℓ_2 then Work done $W = \frac{1}{2} k(\ell_2^2 - \ell_1^2)$



7.1. Spring Pendulum

SL AL

- (i) When a small mass is suspended from a mass-less spring then this arrangement is known as spring pendulum. For small linear displacement the motion of spring pendulum is simple harmonic.
- (ii) For a spring pendulum

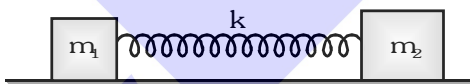
$$F = -kx \Rightarrow m \frac{d^2x}{dt^2} = -kx \quad [\because F = ma = m \frac{d^2x}{dt^2}]$$

$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{k}{m}x \therefore \frac{d^2x}{dt^2} = -\omega^2x \Rightarrow \omega^2 = \frac{k}{m}$$

This is standard equation of linear S.H.M.

$$\text{Time period } T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}, \text{ Frequency } n = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

- (iii) Time period of a spring pendulum is independent of acceleration due to gravity. This is why a clock based on oscillation of spring pendulum will keep proper time everywhere on a hill or moon or in a satellite or different places of earth.
- (iv) If a spring pendulum oscillates in a vertical plane is made to oscillate on a horizontal surface or on an inclined plane then time period will remain unchanged.
- (v) By increasing the mass, time period of spring pendulum increases ($T \propto \sqrt{m}$), but by increasing the force constant of spring (k), Its time period decreases $\left[T \propto \frac{1}{\sqrt{k}}\right]$ whereas frequency increases ($n \propto \sqrt{k}$)
- (vi) If two masses m_1 and m_2 are connected by a spring and made to oscillate then time period $T = 2\pi\sqrt{\frac{\mu}{k}}$

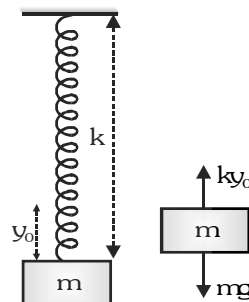


$$\text{Here, } \mu = \frac{m_1 m_2}{m_1 + m_2} = \text{reduced mass}$$

- (vii) If the stretch in a vertically loaded spring is y_0 then for equilibrium of mass m .

$$ky_0 = mg \quad \text{i.e., } \frac{m}{k} = \frac{y_0}{g}$$

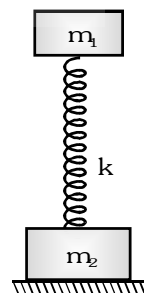
$$\text{So, time period } T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{y_0}{g}}$$



But remember time period of spring pendulum is independent of acceleration due to gravity.

- (viii) If two particles are attached with spring in which only one is oscillating

$$\text{Time period} = 2\pi\sqrt{\frac{\text{mass of oscillating particle}}{\text{force constant}}} = 2\pi\sqrt{\frac{m_1}{k}}$$

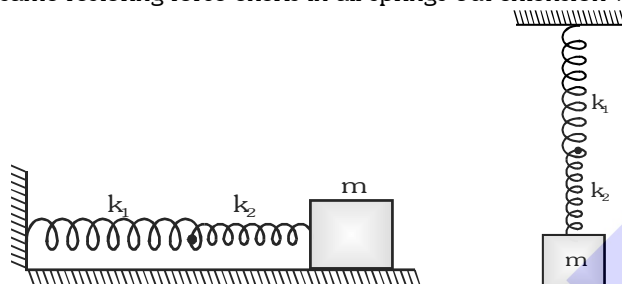


7.2. Various Spring arrangements

SL AL

Series combination of springs

In series combination same restoring force exerts in all springs but extension will be different.



Total displacement $x = x_1 + x_2$

Force acting on both springs $F = -k_1 x_1 = -k_2 x_2$

$$\therefore x_1 = -\frac{F}{k_1} \text{ and } x_2 = -\frac{F}{k_2} \quad \therefore x = -\left[\frac{F}{k_1} + \frac{F}{k_2}\right] \quad \dots(i)$$

If equivalent force constant is k_s then $F = -k_s x$

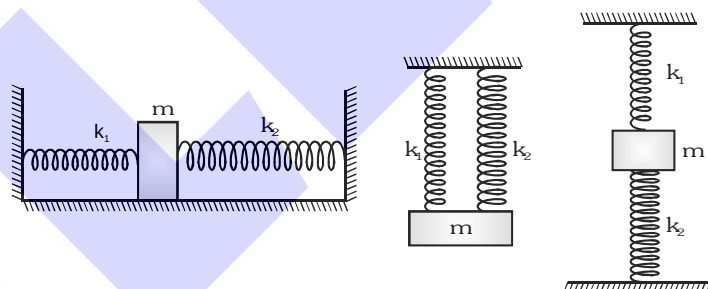
$$\text{so by equation (i)} \quad -\frac{F}{k_s} = -\frac{F}{k_1} - \frac{F}{k_2} \Rightarrow \frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2} \Rightarrow k_s = \frac{k_1 k_2}{k_1 + k_2}$$

$$\text{Time period } T = 2\pi\sqrt{\frac{m}{k_s}} = 2\pi\sqrt{\frac{m(k_1 + k_2)}{k_1 k_2}}$$

$$\text{Frequency } n = \frac{1}{2\pi}\sqrt{\frac{k_s}{m}}, \text{ Angular frequency } \omega = \sqrt{\frac{k_s}{m}}$$

7.3. Parallel Combination of springs

SL AL



In parallel combination displacement on each spring is same but restoring force is different.

Force acting on the system $F = F_1 + F_2 \Rightarrow F = -k_1 x - k_2 x \dots(i)$

If equivalent force constant is k_p then, $F = -k_p x$, so by equation (i) $-k_p x = -k_1 x - k_2 x \Rightarrow k_p = k_1 + k_2$

$$\text{Time period } T = 2\pi\sqrt{\frac{m}{k_p}} = 2\pi\sqrt{\frac{m}{k_1 + k_2}}; \text{ Frequency } n = \frac{1}{2\pi}\sqrt{\frac{k_p}{m}}; \text{ Angular frequency } \omega = \sqrt{\frac{k_1 + k_2}{m}}$$

Illustrations

Illustration 14. A body of mass m attached to a spring which is oscillating with time period 4 seconds. If the mass of the body is increased by 4 kg, its time period increases by 2 sec. Determine value of initial mass m .

Solution In 1st case : $T = 2\pi\sqrt{\frac{m}{k}} \Rightarrow 4 = 2\pi\sqrt{\frac{m}{k}} \dots(i)$ and in 2nd case: $6 = 2\pi\sqrt{\frac{m+4}{k}} \dots(ii)$

Divide (i) by (ii) $\frac{4}{6} = \sqrt{\frac{m}{m+4}} \Rightarrow \frac{16}{36} = \frac{m}{m+4} \Rightarrow m = 3.2 \text{ kg}$

Illustration 15. One body is suspended from a spring of length ℓ , spring constant k and has time period T . Now if spring is divided in two equal parts which are joined in parallel and the same body is suspended from this arrangement then determine new time period.

Solution

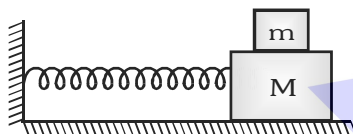
Spring constant in parallel combination $k' = 2k + 2k = 4k$

$$\therefore T' = 2\pi\sqrt{\frac{m}{k'}} = 2\pi\sqrt{\frac{m}{4k}} = 2\pi\sqrt{\frac{m}{k}} \times \frac{1}{\sqrt{4}} = \frac{T}{2}$$

Illustration 16. A block is on a horizontal slab which is moving horizontally and executing S.H.M. The coefficient of static friction between block and slab is μ . If block is not separated from slab then determine angular frequency of oscillation.

Solution

If block is not separated from slab then restoring force due to S.H.M. should be less than frictional force between slab and block.



$$F_{\text{restoring}} \leq F_{\text{friction}} \Rightarrow m a_{\text{max.}} \leq \mu mg \Rightarrow a_{\text{max.}} \leq \mu g \Rightarrow \omega^2 A \leq \mu g \Rightarrow \omega \leq \sqrt{\frac{\mu g}{A}}$$

Illustration 17. A block of mass m is suspended from a spring of spring constant k . Find the amplitude of S.H.M.

Solution

Let amplitude of S.H.M. be x_0 , at mean position

$$Kx_0 = mg$$

$$x_0 = \frac{mg}{K}$$

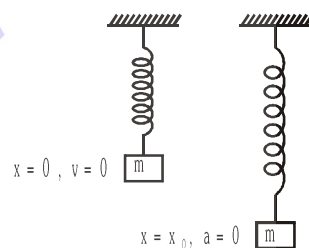


Illustration 18. Periodic time of oscillation T_1 is obtained when a mass is suspended from a spring if another spring is used with same mass then periodic time of oscillation is T_2 . Now if this mass is suspended from series combination of above springs then calculate the time period.

Solution

$$T_1 = 2\pi\sqrt{\frac{m}{k_1}} \Rightarrow T_1^2 = 4\pi^2 \frac{m}{k_1} \Rightarrow k_1 = \frac{4\pi^2 m}{T_1^2} \text{ and}$$

$$T_2 = 2\pi\sqrt{\frac{m}{k_2}} \Rightarrow T_2^2 = 4\pi^2 \frac{m}{k_2} \Rightarrow k_2 = \frac{4\pi^2 m}{T_2^2}$$

$$\text{Now } T = 2\pi\sqrt{\frac{m}{k'}}$$

where

$$\frac{1}{k'} = \frac{1}{k_1} + \frac{1}{k_2} \Rightarrow k' = \frac{k_1 k_2}{k_1 + k_2} = \frac{\left(\frac{4\pi^2 m}{T_1^2}\right) \left(\frac{4\pi^2 m}{T_2^2}\right)}{\frac{4\pi^2 m}{T_1^2} + \frac{4\pi^2 m}{T_2^2}}; k' = \frac{4\pi^2 m \left[\frac{4\pi^2 m}{T_1^2 T_2^2}\right]}{4\pi^2 m \left[\frac{1}{T_1^2} + \frac{1}{T_2^2}\right]} = \frac{4\pi^2 m}{T_1^2 + T_2^2}$$

$$\therefore T = 2\pi\sqrt{\frac{m}{k'}} = 2\pi\sqrt{\frac{m}{\frac{4\pi^2 m}{T_1^2 + T_2^2}}}$$

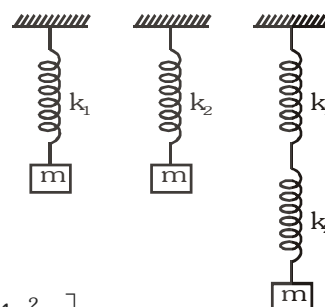


Illustration 19. Infinite spring with force constants $k, 2k, 4k, 8k, \dots$ respectively are connected in series. Calculate the effective force constant of the spring.

Solution
$$\frac{1}{k_{\text{eff}}} = \frac{1}{k} + \frac{1}{2k} + \frac{1}{4k} + \frac{1}{8k} + \dots \infty$$

(For infinite G.P. $S_{\infty} = \frac{a}{1-r}$ where a = First term, r = common ratio)

$$\frac{1}{k_{\text{eff}}} = \frac{1}{k} \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right] = \frac{1}{k} \left[\frac{1}{1 - \frac{1}{2}} \right] = \frac{2}{k} \quad \text{so } k_{\text{eff}} = k/2$$

Illustration 20. Figure shows a system consisting of a massless pulley, a spring of force constant $k = 4000 \text{ N/m}$ and a block of mass $m = 1 \text{ kg}$. If the block is slightly displaced vertically down from its equilibrium position and released find the frequency of its vertical oscillation in given cases.

Solution

Case (A) :

As the pulley is fixed and string is inextensible, if mass m is displaced by y the spring will stretch by y , and as there is no mass between string and spring (as pulley is massless) $F = T = ky$ i.e., restoring force is linear and so motion of mass m will be linear simple harmonic with frequency

$$n_A = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{4000}{1}} \approx 10 \text{ Hz}$$

Case (B) :

The pulley is movable and string inextensible, so if mass m moves down a distance y , the pulley will move down by $(y/2)$. So the force in the spring $F = k(y/2)$.

Now as pulley is massless $F = 2T$, $\Rightarrow T = F/2 = (k/4)y$. So the restoring

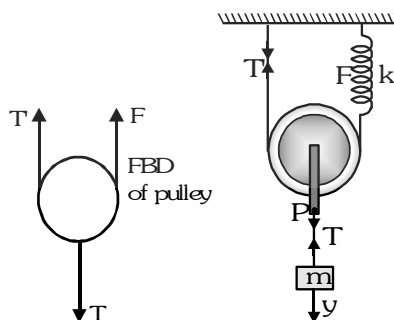
force on the mass m $T = \frac{1}{4}ky = k'y \Rightarrow k' = \frac{1}{4}k$

$$\text{So } n_B = \frac{1}{2\pi} \sqrt{\frac{k'}{m}} = \frac{1}{2\pi} \sqrt{\frac{k}{4m}} = \frac{n_A}{2} = 5 \text{ Hz}$$

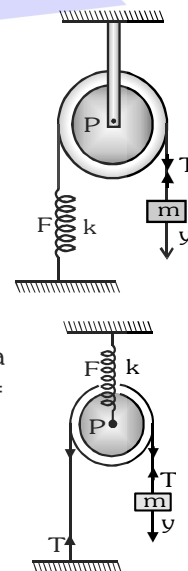
Case (C) :

In this situation if the mass m moves by y the pulley will also move by y and so the spring will stretch by $2y$ (as string is inextensible) and so $T' = F = 2ky$. Now as pulley is massless so $T = F + T' = 4ky$, i.e., the restoring force on the mass m

$T = 4ky = k'y \Rightarrow k' = 4k$



$$\text{so } n_C = \frac{1}{2\pi} \sqrt{\frac{k'}{m}} = \frac{1}{2\pi} \sqrt{\frac{4k}{m}} = 2n_A = 20 \text{ Hz}$$



BEGINNER'S BOX-3

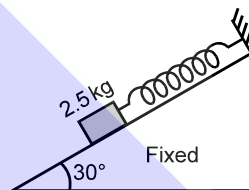
Spring System in SHM

1. A toy car of mass m is having two similar rubber ribbons attached to it as shown in the figure. The force constant of rubber ribbons is k . The car is displaced from mean position by x cm and released. At the mean position the ribbons are undeformed. Vibration period is



- (A) $2\pi\sqrt{\frac{m(2k)}{k^2}}$ (B) $\frac{1}{2\pi}\sqrt{\frac{m(2k)}{k^2}}$ (C) $2\pi\sqrt{\frac{m}{k}}$ (D) $2\pi\sqrt{\frac{m}{k+k}}$

2. A smooth inclined plane having angle of inclination 30° with horizontal has a mass 2.5 kg held by a spring which is fixed at the upper end. If the mass is taken 2.5 cm up along the surface of the inclined plane, the tension in the spring reduces to zero. If the mass is then released, the angular frequency of oscillation in radian per second is

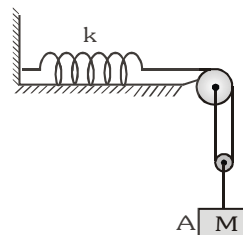


- (A) 0.707 (B) 7.07
(C) 1.414 (D) 14.14

3. A ball of mass m kg hangs from a spring of spring constant k . The ball oscillates with a period of T seconds. If the ball is removed, the spring is shortened by

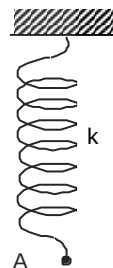
- (A) $\frac{gT^2}{(2\pi)^2}$ meter (B) $\frac{3T^2g}{(2\pi)^2}$ meter (C) $\frac{Tm}{k}$ meter (D) $\frac{Tk}{m}$ meter

4. Block A in the figure is released from the rest when the extension in the spring is x_0 . The maximum downward displacement of the block.



- (A) $\frac{Mg}{2k} - x_0$ (B) $\frac{Mg}{2k} + x_0$
(C) $\frac{2Mg}{k} - x_0$ (D) $\frac{2Mg}{k} + x_0$

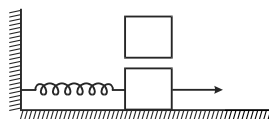
5. A light spring of stiffness k is hanging from the ceiling. It is stretched by a distance d , then a mass m attached to the end A which is then released.



The amplitude of the oscillations will be

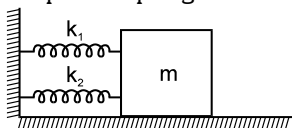
- (A) mg/k (B) $2mg/k$
(C) d (D) $|(mg/k) - d|$

6. A 1 kg block is executing simple harmonic motion of amplitude 0.1 m on a smooth horizontal surface under the restoring force of a spring of spring constant 100 N/m. A block of mass 3 kg is gently placed on it at the instant it passes through the mean position. Assuming that the two blocks move together, find the frequency and the amplitude of the motion.

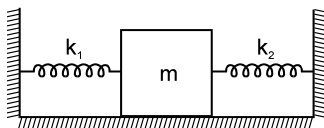


7. A spring stores 5 J of energy when stretched by 25 cm. It is kept vertical with the lower end fixed. A block fastened to its other end is made to undergo small oscillations. If the block makes 5 oscillations each second, what is the mass of the block?

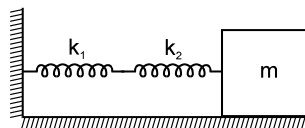
8. Find the time period of the oscillation of mass m in figures a, b, c. What is the equivalent spring constant of the pair of springs in each case?



(a)



(b)



(c)

8.0 SIMPLE PENDULUM

SL AL

If a heavy point mass is suspended by a weightless, inextensible and perfectly flexible string from a rigid support, then this arrangement is called a simple pendulum

8.1. Expression for time period

SL AL

Restoring force acting on pendulum $F = -mg \sin \theta$

For small angle $\sin \theta \approx \frac{OA}{SA} = \frac{y}{\ell} \therefore ma = -mg \times \frac{y}{\ell} \Rightarrow a = -\frac{g}{\ell} y$

It proves that if displacement is small then simple pendulum performs S.H.M.

$$\therefore |a| = \omega^2 y \Rightarrow \omega^2 = \frac{g}{\ell}$$

$$\Rightarrow \omega = \sqrt{\frac{g}{\ell}} \therefore T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{\ell}{g}} = 2\pi \sqrt{\frac{\text{displacement}}{\text{acceleration}}}$$

- $T = 2\pi \sqrt{\frac{\ell}{g}}$ is valid when length of simple pendulum (ℓ) is negligible as compare to radius of earth

($\ell \ll R_e$) but if ℓ is comparable to radius of earth then time period $T = 2\pi \sqrt{\frac{R_e}{\left[1 + \frac{R_e}{\ell}\right] g}}$

- The time period of oscillation of simple pendulum of infinite length ($\ell \rightarrow \infty$)

$$T = 2\pi \sqrt{\frac{R}{g}} \approx 84.6 \text{ minute} \approx 1 \frac{1}{2} \text{ hour (It is maximum time period)}$$

- If angular amplitude (θ_0) is large ($\theta_0 > 15^\circ$) then time period is given by $T = 2\pi \sqrt{\frac{\ell}{g} \left[1 + \frac{\theta_0^2}{16}\right]}$ here θ_0 is in radian.
- If a simple pendulum of density ρ is made to oscillate in a liquid of density σ then its time period will increase as compare to that of air and is given by $T = 2\pi \sqrt{\frac{\ell}{\left[1 - \frac{\sigma}{\rho}\right] g}}$

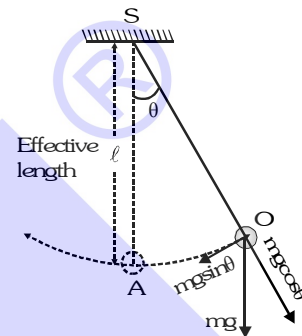
8.2. Second's pendulum

SL AL

If the time period of a simple pendulum is 2 second then it is called second's pendulum. Second's pendulum take one second to go from one extreme position to other extreme position.

For second's pendulum, time period $T = 2 = 2\pi \sqrt{\frac{\ell}{g}}$. At the surface of earth $g = 9.8 \text{ m/s}^2 \approx \pi^2 \text{ m/s}^2$.

So length of second pendulum at the surface of earth $\ell \approx 1 \text{ meter}$



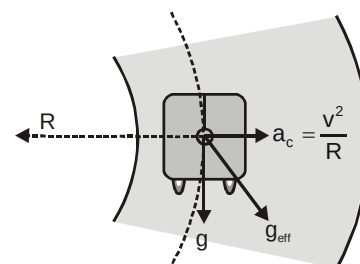
Illustrations

Illustration 21. A simple pendulum of length L and mass M is suspended in a car. The car is moving on a circular track of radius R with a uniform speed v . If the pendulum makes oscillation in a radial direction about its equilibrium position, then calculate its time period.

Solution

Centripetal acceleration $a_c = \frac{v^2}{R}$

& Acceleration due to gravity = g ;



$$\text{So } g_{\text{eff}} = \sqrt{g^2 + \left(\frac{v^2}{R}\right)^2}$$

$$\Rightarrow \text{Time period } T = 2\pi\sqrt{\frac{L}{g_{\text{eff}}}} = 2\pi\sqrt{\frac{L}{\sqrt{g^2 + \frac{v^4}{R^2}}}}$$

Illustration 22. A simple pendulum is suspended from the ceiling of a lift. When the lift is at rest, its time period is T . With what acceleration should lift be accelerated upwards in order to reduce its time period to $\frac{T}{2}$.

Solution In stationary lift $T = 2\pi\sqrt{\frac{\ell}{g}}$... (i)

In accelerated lift $\frac{T}{2} = T' = 2\pi\sqrt{\frac{\ell}{g+a}}$... (ii)

Divide (i) by (ii) $2 = \sqrt{\frac{g+a}{g}} \Rightarrow g+a = 4g \Rightarrow a = 3g$

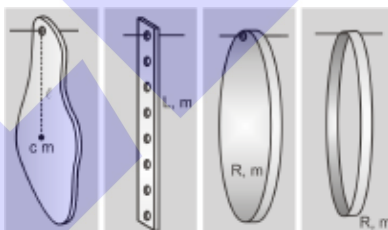
9.0 COMPOUND PENDULUM

SL AL

Any rigid body which is free to oscillate in a vertical plane about a horizontal axis passing through a point, is define compound pendulum

9.1. Expression for time period

SL AL



Torque acting on a body $\tau = -mg\ell\sin\theta$ if angle is very small $\sin\theta \approx \theta$ then

$$\tau = -mg\ell\theta \quad \dots(i)$$

$$\text{and } \tau = I_s \alpha \quad \dots(ii)$$

Here m = mass of the body

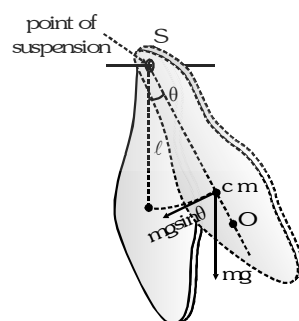
ℓ = distance between point of suspension and centre of mass

I_s = moment of inertia about horizontal axis passes through point of suspension

From equation (i) and (ii) $I_s \alpha = -mg\ell\theta$

$$I_s \frac{d^2\theta}{dt^2} + mg\ell\theta = 0, \quad \frac{d^2\theta}{dt^2} + \frac{mg\ell}{I_s} \theta = 0 \quad \dots(iii)$$

$$\therefore \frac{d^2\theta}{dt^2} + \omega^2\theta = 0 \quad \dots(iv)$$



Compare equation (iii) and (iv) $\omega^2 = \frac{mg\ell}{I_s} \Rightarrow \omega = \sqrt{\frac{mg\ell}{I_s}}$

Time period of compound pendulum $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I_s}{mg\ell}}$

Applying parallel axis theorem $I_s = I_{CM} + m\ell^2 \Rightarrow I_s = mK^2 + m\ell^2$

$$\therefore T = 2\pi\sqrt{\frac{I_s}{mg\ell}} = 2\pi\sqrt{\frac{mK^2 + m\ell^2}{mg\ell}} \Rightarrow T = 2\pi\sqrt{\frac{K^2}{\ell} + \ell \over g}$$

Here S = point of suspension ; O = point of oscillation ; K = radius of gyration

$$L = 0 \frac{K^2}{\ell} + \ell = \text{equivalent length of simple pendulum}$$

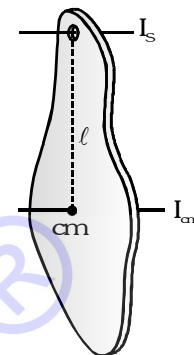
ℓ = distance between point of suspension and point of oscillation

$$\text{Time Period } T = 2\pi\sqrt{\frac{K^2}{\ell} + \ell \over g}$$

For maximum time period $\ell = 0$ Maximum time period $T_{\max} = \infty$

For minimum time period $\frac{dT}{d\ell} = 0$ then $K = \ell \Rightarrow T_{\min} = T = 2\pi\sqrt{\frac{K^2}{K} + K \over g} = 2\pi\sqrt{\frac{2K}{g}}$

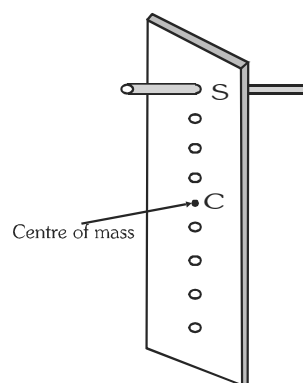
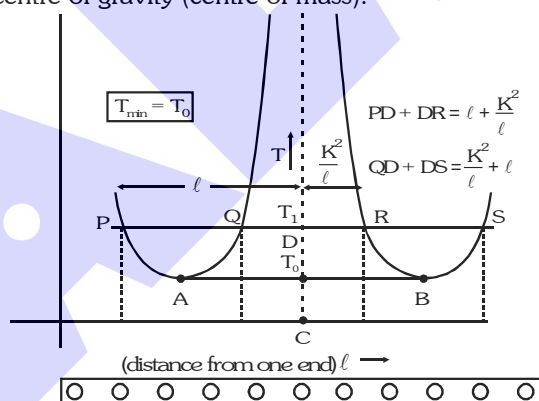
Note : $\frac{K^2}{\ell}$ = Distance between centre of mass and point of oscillation. It has uniform cross section area and width at any point.



9.2. Bar pendulum

SL AL

A bar pendulum is a steel bar of 1 meter length with holes at regular intervals for suspension. The time period is measured for different values of ℓ (distance between S and C). The graph between T and length from one end ℓ is as shown in fig below. The time period is infinite when $\ell = 0$, i.e., when it is suspended from the centre of gravity (centre of mass).



At four points P, Q, R and S, the time period is the same T_1 .

The distance are such that

$$\frac{PR + QS}{2} = \ell_{eq} = \ell + K^2/\ell \quad \text{Also} \quad PD = \ell \quad DR = K^2/\ell$$

The time period is minimum when $\ell = K$

In Fig. $AB = 2K$. The minimum period is $T_0 = 2\pi\sqrt{\frac{2K}{g}}$

Condition for T minimum

$$T = \frac{2\pi}{g} \left(\frac{K^2}{\ell} + \ell \right)$$

diff. w.r. to ℓ

$$2T \frac{dT}{d\ell} = \left(\frac{4\pi^2}{g} \right) \left[-\frac{K^2}{\ell^2} + 1 \right]$$

$$\therefore T \neq 0, \text{ and with } \frac{dT}{d\ell} = 0$$

$$-\frac{K^2}{\ell^2} + 1 = 0 \text{ or } K^2 = \ell^2$$

$$K = \pm \ell \text{ then } T_{\min} = T_0$$

Illustrations

Illustration 23. A disc is made to oscillate about a horizontal axis passing through mid-point of its radius. Determine time period.

Solution

$$\text{For disc } I = MK^2 = \frac{MR^2}{2} \Rightarrow K = \frac{R}{\sqrt{2}}, \ell = \frac{R}{2}$$

$$L = \ell + \frac{K^2}{\ell} = \frac{R}{2} + \frac{R^2}{2\left(\frac{R}{2}\right)} = \frac{R}{2} + R = \frac{3R}{2}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{L}{g}} = 2\pi\sqrt{\frac{3R}{2g}}$$

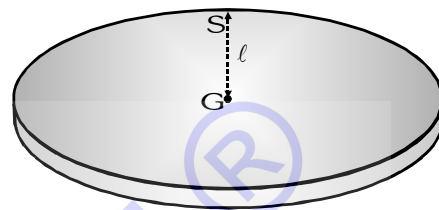


Illustration 24. A rod with rectangular cross section oscillates about a horizontal axis passing through one of its ends and it behaves like a second's pendulum. Determine its length.

Solution Because oscillating rod behaves as a second's pendulum so its time period will be 2 second.

$$T = 2\pi\sqrt{\frac{\ell + \frac{K^2}{\ell}}{g}} = 2s \Rightarrow \ell + \frac{K^2}{\ell} = 1 \quad \dots(i) \quad [\because \pi^2 = g]$$

Assume length of rod is L, because axis passes through one end

So $\ell = \frac{L}{2}$ and $K^2 = \frac{L^2}{12}$

Putting this values in equation we get $\frac{L}{2} + \frac{L^2}{12} \times \frac{2}{L} = 1 \Rightarrow L = 1.5 \text{ m}$

Illustration 25. The time period of a bar pendulum when suspended at distances 30 cm and 50 cm from its centre of gravity comes out to be the same. If the mass of the body is 2kg. Find out its moment of inertia about an axis passing through first point.

Solution

$$I_A = MK^2 + M\ell_1^2 \text{ but } \ell_1 = 50 \text{ cm \& } \frac{K^2}{\ell_1} = 30 \text{ cm}$$

$$\Rightarrow K^2 = 30 \times 50 \text{ cm}^2$$

$$I_A = 2 \times (30 \times 50 \times 10^{-4}) + 2 \times (50 \times 10^{-2})^2$$

$$= 0.3 + 0.18 = 0.48 \text{ kg-m}^2$$

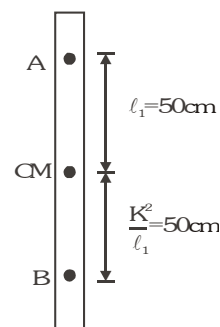


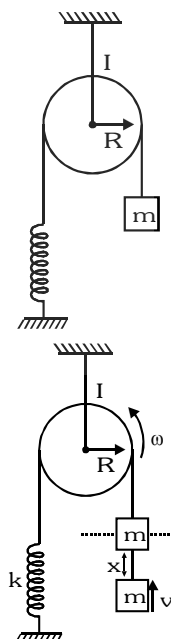
Illustration 26. Figure shows a pulley block system in equilibrium. If the block is displaced down slightly from its equilibrium position and released. Find the time period of oscillation of the system. Assume there is sufficient friction present between pulley and string so that string will not slip over pulley surface.

Solution If m is in equilibrium tension in string must be mg and spring is stretched by h so that $mg = kh$. If we displace the block downward by a distance A and released, it starts executing SHM with amplitude A. During its oscillation we consider the block at a displacement x below the equilibrium position, if it is moving at a speed v at this position, the pulley will be rotating at an angular speed ω given as $\omega = \frac{v}{r}$

Thus at this position the total energy of oscillating system is

$$E_T = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + \frac{1}{2}k(x+h)^2 - mgx$$

Differentiating with respect to time, we get



$$\frac{dE_T}{dt} = \frac{1}{2}m \left(2v \frac{dv}{dt} \right) + \frac{1}{2}I \left(\frac{1}{r^2} \right) \left(2v \frac{dv}{dt} \right) + \frac{1}{2}k \left[2(x+h) \frac{dx}{dt} \right] - mg \left(\frac{dx}{dt} \right) = 0$$

$$\text{or } mva + \frac{1}{r^2}va + k(x+h)v - mgv = 0 \text{ or } a + \left(\frac{k}{m + \frac{1}{r^2}} \right)x = 0 \text{ [as } mg = kv]$$

Comparing above equation with standard differential equation of SHM we get

$$\omega = \sqrt{\frac{k}{m + \frac{1}{r^2}}}$$

$$\text{Thus time period of oscillation is } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m + \frac{1}{r^2}}{k}}$$

Illustration 27*. Figure shows a pulley block system in which a block A is hanging on one side of pulley and an other side a small bead B of mass m is welded on pulley. The moment of inertia of pulley is I and the system is in equilibrium when bead is at an angle α from the vertical. If the system is slightly disturbed from its equilibrium position, find the time period of its oscillations.

Solution

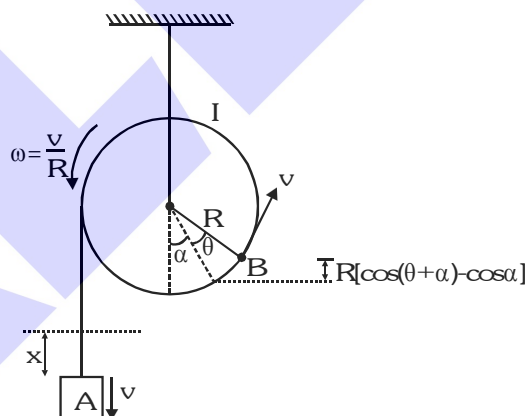
In equilibrium the net torque on pulley must be zero, thus we have

$$MgR = mgR \sin \alpha \text{ or } M = m \sin \alpha$$

[if mass of block A is assumed to be M]

Now if block is displaced down by distance A and released, it starts oscillating with amplitude A . Now consider the block at a distance x below the equilibrium position when it is going down at speed v . Figure shows the corresponding situation at this instant and the total energy of oscillating system can be written as

$$E_T = \frac{1}{2}Mv^2 + \frac{1}{2}mv^2 + \frac{1}{2}I \left(\frac{v}{R} \right)^2 - Mgx + mgR[\cos \alpha - \cos(\theta + \alpha)]$$

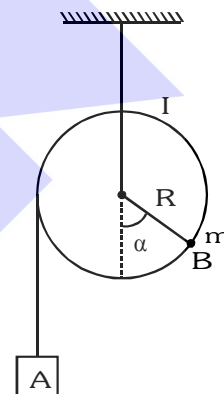


Differentiating the above equation w.r. to time, we get

$$\frac{dE_T}{dt} = \frac{1}{2}M \left(2v \frac{dv}{dt} \right) + \frac{1}{2}m \left(2v \frac{dv}{dt} \right) + \frac{1}{2} \frac{I}{R^2} \left(2v \frac{dv}{dt} \right) - Mg \left(\frac{dx}{dt} \right) + mgR \left[-\sin(\theta + \alpha) \frac{d\theta}{dt} \right] = 0$$

$$\Rightarrow Mva + mva + \frac{I}{R^2}va - Mgv + mgR \sin(\theta + \alpha) \left(\frac{v}{R} \right) = 0 \left[\text{as } \frac{d\theta}{dt} = \omega = \frac{v}{R} \right]$$

$$\Rightarrow \left(M + m + \frac{I}{R^2} \right) a - Mg - mg[\theta \cos \alpha + \sin \alpha] = 0$$



$$\Rightarrow \left(M + m + \frac{I}{R^2} \right) a + mg \cos \alpha \cdot \frac{x}{R} = 0 \quad \left[\text{as } M = m \sin \alpha \text{ and } \theta = \frac{x}{R} \right] \Rightarrow a = - \frac{mg \cos \alpha}{R \left(M + m + \frac{I}{R^2} \right)} x$$

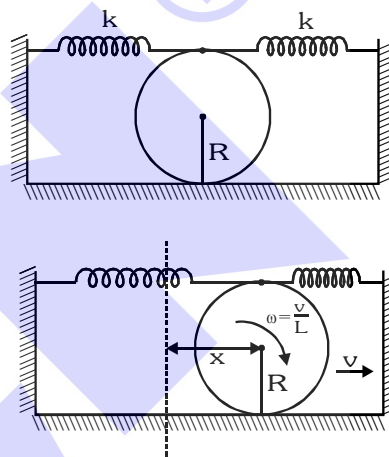
Comparing equation with basic differential equation of SHM, we get the angular frequency of

$$\text{SHM of system as } \omega = \sqrt{\frac{mg \cos \alpha}{R \left(M + m + \frac{I}{R^2} \right)}}$$

Illustration 28*. A solid uniform cylinder of mass M performs small oscillations in horizontal plane if slightly displaced from its mean position shown in figure. If it is given that initially springs are in natural lengths and cylinder does not slip on ground during oscillations due to friction between ground and cylinder. Force constant of each spring is k . Find time period of these oscillation.

Solution

In the situation given in problem, the cylinder is in its equilibrium position when springs are unstrained. When it is slightly rolled and released. It starts executing SHM and due to friction, the cylinder is in pure rolling motion. Now during oscillations we consider the cylinder when it is at a distance x from the mean position and moving with a speed v as shown in figure. As cylinder is in pure rolling, its angular speed of rotation can be given as



As centre of cylinder is at a distance x from the initial position, the springs which are connect at a point on its rim must be compressed and stretched by a distance $2x$. Thus at this intermediate position total energy of the oscillating system can be given as

$$E_T = \frac{1}{2} M v^2 + \frac{1}{2} \left(\frac{1}{2} M R^2 \right) \omega^2 + \frac{1}{2} k (2x)^2 \times 2$$

Differentiating with respect to time, we get

$$\frac{dE_T}{dt} = \frac{1}{2} M \left(2v \frac{dv}{dt} \right) + \frac{1}{2} \left(\frac{1}{2} M R^2 \right) \frac{1}{R^2} \left(2v \frac{dv}{dt} \right) + 4k \left(2x \frac{dx}{dt} \right) = 0$$

$$\Rightarrow M v a + \frac{1}{2} M v a + 8k x v = 0 \Rightarrow a = - \frac{16k}{3M} x$$

Comparing equation with basic differential equation of SHM.

$$\text{We get, the angular frequency of SHM as } \omega = \sqrt{\frac{16k}{3M}}$$

$$\text{Thus time period of these oscillation is } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{3M}{16k}} = \frac{\pi}{2} \sqrt{\frac{3M}{k}}$$

BEGINNER'S BOX-4

Simple Pendulum and Compound Pendulum

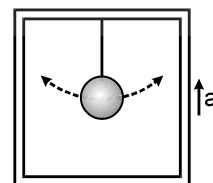
1. A scientist measures the time period of a simple pendulum as T in a lift at rest. If the lift moves up with acceleration as one fourth of the acceleration of gravity, the new time period is

(A) $\frac{T}{4}$

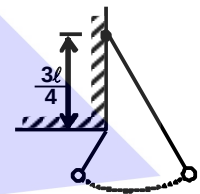
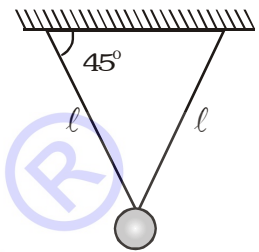
(B) $4T$

(C) $\frac{2}{\sqrt{5}} T$

(D) $\frac{\sqrt{5}}{2} T$



2. The motion of a torsional pendulum is
(A) periodic (B) oscillatory
(C) simple harmonic (D) angular simple harmonic
3. Time period of small oscillation (in a vertical plane normal to the plane of strings) of the bob in the arrangement shown will be
(A) $2\pi\sqrt{\frac{\ell}{g}}$ (B) $2\pi\sqrt{\frac{\ell}{\sqrt{2}g}}$
(C) $2\pi\sqrt{\frac{\sqrt{2}\ell}{g}}$ (D) $2\pi\sqrt{\frac{2\ell}{g}}$
4. A simple pendulum of length ℓ has a time period T for small oscillations. A fixed obstacle is placed directly below the point of suspension, so that only the lower quarter of the string continues oscillations (see figure). The pendulum is released from rest at a certain point O. The period of oscillation, assuming small angles, will be:
(A) $3T/4$ (B) $T/4$
(C) $T/2 (1 + \sqrt{3})$ (D) T
5. A uniform disc of radius r is to be suspended through a small hole made in the disc. Find the minimum possible time period of the disc for small oscillations. What should be the distance of the hole from the centre for it to have minimum time period?



10.0 SUPERPOSITION OF TWO SHM'S

10.1 In same direction and of same frequency

SL AL

$$x_1 = A_1 \sin \omega t$$

$$x_2 = A_2 \sin (\omega t + \theta), \text{ then resultant displacement}$$

$$x = x_1 + x_2 = A_1 \sin \omega t + A_2 \sin (\omega t + \theta) = A \sin (\omega t + \phi)$$

$$\text{where } A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \theta}$$

$$\& \quad \phi = \tan^{-1} \left[\frac{A_2 \sin \theta}{A_1 + A_2 \cos \theta} \right]$$

$$\text{If } \theta = 0, \text{ both SHM's are in phase and } A = A_1 + A_2$$

$$\text{If } \theta = \pi, \text{ both SHM's are out of phase and } A = |A_1 - A_2|$$

The resultant amplitude due to superposition of two or more than two SHM's of this case can also be found by phasor diagram also.

10.2 In same direction but are of different frequencies.

SL AL

$$x_1 = A_1 \sin \omega_1 t; \quad x_2 = A_2 \sin \omega_2 t$$

$$\text{then resultant displacement } x = x_1 + x_2 = A_1 \sin \omega_1 t + A_2 \sin \omega_2 t$$

This resultant motion is not SHM.

10.3 In two perpendicular directions

SL AL

$$x = A \sin \omega t$$

$$y = B \sin (\omega t + \theta)$$

Case (i) : If $\theta = 0$ or π then $y = \pm (B/A) x$. So path will be straight line & resultant displacement will be $r = (x^2 + y^2)^{1/2} = (A^2 + B^2)^{1/2} \sin \omega t$

which is equation of SHM having amplitude $\sqrt{A^2 + B^2}$

Case (ii) : If $\theta = \frac{\pi}{2}$ then $x = A \sin \omega t$

$$y = B \sin (\omega t + \pi/2) = B \cos \omega t$$

so, resultant will be $\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$. i.e. equation of an ellipse and if $A = B$, then superposition will be an equation of circle.

10.4 Superposition of SHM's along the same direction (using phasor diagram)

SL AL

If two or more SHM's are along the same line, their resultant can be obtained by vector addition by making phasor diagram.

1. Amplitude of SHM is taken as length(magnitude) of vector.
2. Phase difference between the vectors is taken as the angle between these vectors. The magnitude of resultant of vector's give resultant amplitude of SHM and angle of resultant vector gives phase constant of resultant SHM.

For example;

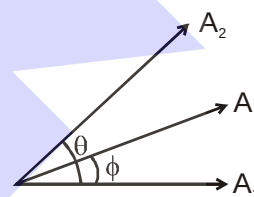
$$x_1 = A_1 \sin \omega t$$

$$x_2 = A_2 \sin (\omega t + \theta)$$

If equation of resultant SHM is taken as $x = A \sin (\omega t + \phi)$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \theta}$$

$$\tan \phi = \frac{A_2 \sin \theta}{A_1 + A_2 \cos \theta}$$



Illustrations

Illustration 29. Find the amplitude of the simple harmonic motion obtained by combining the motions

$$x_1 = (2.0 \text{ cm}) \sin \omega t \text{ and } x_2 = (2.0 \text{ cm}) \sin (\omega t + \pi/3).$$

Solution

The two equations given represent simple harmonic motions along X-axis with amplitudes $A_1 = 2.0 \text{ cm}$ and $A_2 = 2.0 \text{ cm}$. The phase difference between the two simple harmonic motions is $\pi/3$. The resultant simple harmonic motion will have an amplitude A given by

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \delta}$$

$$= \sqrt{(2.0 \text{ cm})^2 + (2.0 \text{ cm})^2 + 2(2.0 \text{ cm})^2 \cos \frac{\pi}{3}} = 3.5 \text{ cm}$$

11.0 EXAMPLES OF SIMPLE HARMONIC MOTION

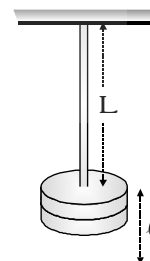
SL AL

If a mass m is suspended from a wire of length L , cross section A and young's modulus Y and is pulled along the length of the wire then restoring force will be developed by the elasticity of the wire.

$$Y = \frac{\text{stress}}{\text{strain}} \Rightarrow \frac{F/A}{\ell/L} \Rightarrow \frac{FL}{A} \Rightarrow F = -\frac{YA}{L} \ell$$

Restoring force is linear so motion is linear simple harmonic with force constant

$$k = \frac{YA}{L} \quad \text{i.e., } n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{YA}{mL}}$$



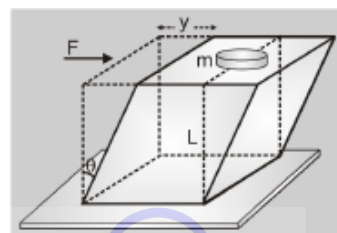
If the lower surface of a cube of side L and of modulus of rigidity η fixed while fixing a particle of mass m on the upper face, a force parallel to upper face is applied and withdrawn; Here restoring force will be developed due to elasticity of block.

Modulus of rigidity of the block $\eta = \frac{\text{shear stress}}{\text{shear strain}}$

$$\eta = \frac{F}{A\theta} \Rightarrow F = \eta \frac{A}{L} y \quad \left[\text{as } \theta = \frac{y}{L} \right]$$

Restoring force is linear so motion will be linear S.H.M.

Force constant $(k) = \eta \frac{A}{L} = \eta L$ [as $A = L^2$] So $T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{m}{\eta L}}$



11.1 Motion of a liquid in a V-shape tube when it is slightly depressed and released

SL AL

Here cross-section of the tube is uniform and the liquid is incompressible and non viscous. Initially the level of liquid in the two limbs will be at the same height. If the liquid is pressed by y in one limb, it will rise by y along the length of the tube in the other limb so the restoring force will developed by hydrostatic pressure difference, i.e.,

$$F = -\Delta P \times A = -(h_1 + h_2)gdA$$

$$\Rightarrow F = -Agd(\sin\theta_1 + \sin\theta_2)y$$

As the restoring force is linear, motion will be linear simple harmonic.

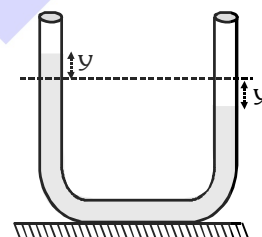
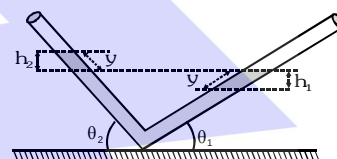
Force constant $(k) = Agd(\sin\theta_1 + \sin\theta_2)$ So $T = 2\pi \sqrt{\frac{m}{Adg(\sin\theta_1 + \sin\theta_2)}}$

Note

If the tube is a U-tube and liquid is filled to a height h

$$\theta_1 = \theta_2 = 90^\circ \text{ and } m = hAd \times 2$$

So time period $T = 2\pi \sqrt{\frac{h}{g}}$



11.2 When a partially submerged floating body is slightly pressed and released

SL AL

If a body of mass m and cross section A is floating in a liquid of density σ with height h inside the liquid then

$$mg = \text{Thrust} = Ah\sigma g, \quad \text{i.e., } m = Ah\sigma \dots(i)$$

Now from this equilibrium position if it is pressed by y , restoring force will developed due to extra thrust i.e.

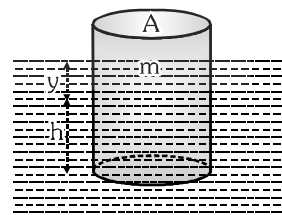
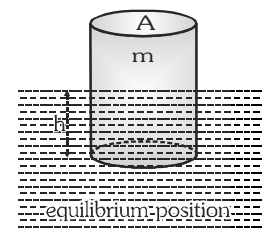
$$F = -A\sigma g y$$

As restoring force is linear, motion will be linear simple harmonic with force constant $k = A\sigma g$.

$$\text{So } T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{m}{A\sigma g}}$$

From this expression it is clear that if density of liquid decreases, time period will increase and vice-versa.

And also as from eqⁿ. (i) $m = Ah\sigma$, $T = 2\pi \sqrt{\frac{h}{g}}$ where h is the height of the body inside the liquid.



11.3 Motion of a ball in a bowl

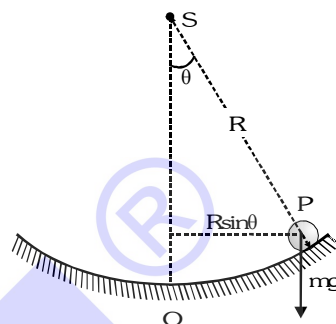
SL AL

If a small steel ball of mass m is placed at a small distance from O inside a smooth concave surface of radius R and released, it will oscillate about O . The restoring torque here will be due to the force of gravity mg on the ball i.e., $\tau = -mg(R\sin\theta) = -mgR\theta$ [As θ is small]

Now as restoring torque is angular so motion will be angular simple harmonic. And as by definition.

$$\tau = I\alpha = mR^2 \left[\frac{d^2\theta}{dt^2} \right] \quad [\text{as } I = mR^2 \text{ and } \alpha = \frac{d^2\theta}{dt^2}]$$

$$\text{so, } mR^2 \frac{d^2\theta}{dt^2} = -mgR\theta \quad \text{i.e., } \frac{d^2\theta}{dt^2} = -\omega^2\theta \Rightarrow \omega^2 = \frac{g}{R} \quad \text{so } T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{R}{g}}$$



11.4 Motion of a ball in a tunnel through the earth

SL AL

Case I

If the tunnel is along a diameter and a ball is released from the surface. If the ball at any time is at a distance y from the centre of earth, then the restoring force will act on the ball due to gravitation between ball and earth. But from theory of gravitation we know that force that acts on a particle inside the earth at a distance y from its centre is only due to mass M' of the earth that lies within sphere of radius y . (the portion of the earth that lies outside this sphere does not exert any net force on the particle) so $F = \frac{-GmM'}{y^2}$

$$\text{But as } M = \frac{4}{3}\pi R^3\rho \quad \text{and} \quad M' = \frac{4}{3}\pi y^3\rho, \text{ i.e., } M' = M \left[\frac{y}{R} \right]^3$$

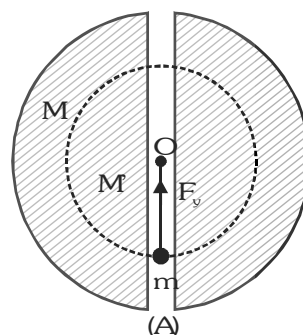
$$F = \frac{-Gm}{y^2} \times M \left[\frac{y^3}{R^3} \right] = -\frac{GMm}{R^3} y$$

Restoring force is linear so the motion is linear SHM with force constant.

$$k = \frac{GMm}{R^3} \quad \text{so } T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{R^3}{GM}}$$

$$\text{Further more as } g = \frac{GM}{R^2} \Rightarrow T = 2\pi\sqrt{\frac{R}{g}}$$

Which is same as that of a simple pendulum of infinite length and is equal to 84.6 minutes.



Case II

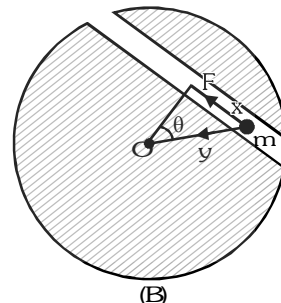
If the tunnel is along a chord and ball is released from the surface and if the ball at any time is at a distance x from the centre of the tunnel. The restoring force will be :

$$F' = F\sin\theta = \left[-\frac{GMm}{R^3} y \right] \left[\frac{x}{y} \right] = -\frac{GMm}{R^3} x$$

Which is again linear with same force constant $k = \frac{GMm}{R^3}$

So that motion is linear simple harmonic with same time period -

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{R^3}{GM}} = 2\pi\sqrt{\frac{R}{g}} = 84.6 \text{ minutes}$$



Note : In SHM $v_{\max} = \omega A$

(i) In I case and IInd case time period will be same but $v_{\max}$ will be different.

(ii) If ball is dropped from height h it will perform oscillatory motion not SHM [$F \propto \frac{1}{r^2}$ and not $F \propto (-r)$]

11.5 Conical Pendulum

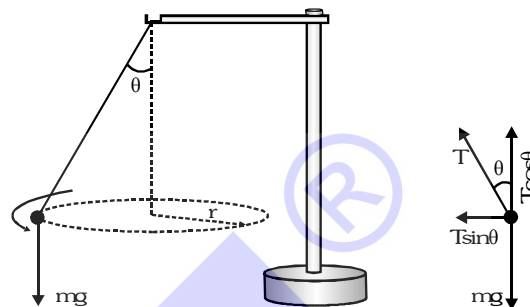
SL AL

It is not Illustration of S.H.M. but Illustration of periodic motion.

$$T = 2\pi\sqrt{\frac{h}{g}} \quad \text{where } h = L \cos\theta$$

$$h = \sqrt{L^2 - r^2}$$

$$\omega = \sqrt{\frac{g}{L \cos\theta}}$$



11.6 Torsional Oscillator (Angular SHM)

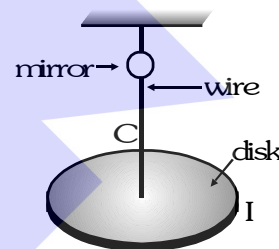
SL AL

$$T = 2\pi\sqrt{\frac{I}{C}}$$

$$\text{where } C = \frac{\eta\pi r^4}{2\ell}$$

η = modulus of elasticity of the wire ; r = radius of the wire

L = length of the wire ; I = Moment of inertia of the disc



11.7 Oscillation of piston in a frictionless gas chamber piston

SL AL

$$T = 2\pi\sqrt{\frac{V_m}{A^2 E}} \quad \text{where } V = \text{volume of cylinder}$$

$$m = \text{mass of piston}, A = \text{area of cylinder ball}, E = \text{bulk modulus} = \frac{\Delta P}{-\Delta V/V}$$

$$\text{For Isothermal process : } E = P, \text{ so } T = 2\pi\sqrt{V_m / PA^2}$$

$$\text{For Adiabatic process : } E = \gamma P, \text{ so } T = 2\pi\sqrt{V_m / \gamma PA^2}$$

Illustrations

Illustration 30. A liquid of mass m is set into oscillations in a U-tube of cross section A . Its time period recorded is T , where $T = 2\pi\sqrt{\frac{\ell}{2g}}$, here ℓ is the length of liquid column. If the liquid of same mass is set into oscillations in U-tube of cross section $\frac{A}{16}$ then determine time period of oscillation.

Solution

Mass is constant $\Rightarrow$ volume $\times$ density = constant $\Rightarrow V_1 d = V_2 d$

$$(A\ell)d = \left[\frac{A}{16}\ell'\right]d$$

$$\Rightarrow \ell' = 16\ell \quad \therefore T = 2\pi\sqrt{\frac{\ell}{2g}}$$

$$\therefore \frac{T'}{T} = \sqrt{\frac{\ell'}{\ell}} = \sqrt{\frac{16\ell}{\ell}} = 4 \quad \Rightarrow T' = 4T$$

Illustration 31. A ball of mass m kept at the centre of a string of length L is pulled from center in perpendicular direction and released. Prove that motion of ball is simple harmonic and determine time period of oscillation

Solution

Restoring force $F = -2T \sin \theta$

When θ is small $\sin \theta \approx \tan \theta \approx \theta = \frac{x}{L/2}$

$$m \frac{d^2 x}{dt^2} = -2T \sin \theta = -2T \theta = -2T \frac{x}{L/2}$$

$$\frac{d^2 x}{dt^2} = -\frac{4T}{mL} x \Rightarrow \frac{d^2 x}{dt^2} \propto -x$$

So motion is simple harmonic $\omega = \frac{2\pi}{T} = \sqrt{\frac{4T}{mL}} \Rightarrow T = 2\pi \sqrt{\frac{mL}{4T}}$

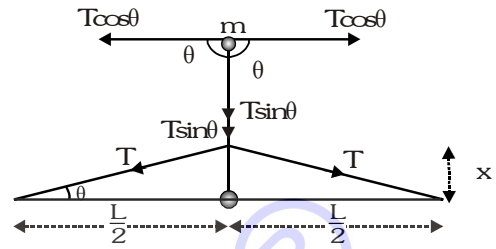
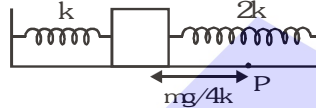


Illustration 32. In the spring mass system shown in the figure, the spring is compressed by $x_0 = \frac{mg}{2k}$ from its natural length and block is released from rest. Find the speed of the block when it passes through P ($mg/4k$ distance from mean position)



Solution

$$\omega = \sqrt{\frac{3k}{m}}, x = A \sin(\omega t + \phi), v = A \omega \cos(\omega t + \phi)$$

$$x = \frac{mg}{4k} = \frac{mg}{2k} \sin(\omega t + \phi)$$

$$\Rightarrow \sin(\omega t + \phi) = \frac{1}{2}$$

$$v = A \omega \cos(\omega t + \phi) = \frac{mg}{2k} \sqrt{\frac{3k}{m}} \times \frac{\sqrt{3}}{2} = g \sqrt{\frac{9m}{16k}}$$

Illustration 33*. A very light rod of length ℓ pivoted at O is connected with two springs of stiffness k_1 & k_2 at a distance of a & ℓ from the pivot respectively. A block of mass m attached with the spring k_2 is kept on a smooth horizontal surface. Find the angular frequency of small oscillation of the block m .

Solution

Let the block be pulled towards right through a distance x ,

then $x = x_B + x_{CB}$

where, x_{CB} = displacement of C (the block) relative to B

$$\text{Thus } x_{CB} = \frac{F}{k_2} \dots \text{(ii)} \quad \text{and} \quad x_B = \left(\frac{F'}{k_1} \right) \frac{\ell}{a} \dots \text{(iii)}$$

Torque acting on the rod about point O,

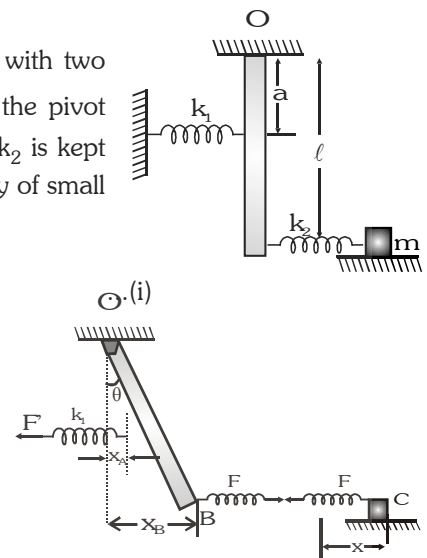
$$\tau_0 = F'a - F\ell \Rightarrow I_0 \frac{d^2 \theta}{dt^2} = F'a - F\ell$$

Since the rod is very light its moment of inertia

$$I_0 \text{ about O is approximately equal to zero} \Rightarrow F' = F \left(\frac{\ell}{a} \right) \dots \text{(iv)}$$

$$\text{Using (iii) \& (iv)} \Rightarrow x_B = \frac{F}{k_1} \left(\frac{\ell}{a} \right)^2 \dots \text{(v)}$$

$$\text{Using (i), (ii) \& (v)} \Rightarrow x = \frac{F}{k_1} \left(\frac{\ell}{a} \right)^2 + \frac{F}{k_2}$$



As force F is opposite to displacement x , then

$$\Rightarrow F = -\frac{k_1 k_2}{k_2 \left(\frac{\ell}{a}\right)^2 + k_1} x$$

$$\Rightarrow m\omega^2 x = \frac{k_1 k_2}{k_2 \left(\frac{\ell}{a}\right)^2 + k_1} x \Rightarrow \omega = \sqrt{\frac{k_1 k_2 a^2}{m(k_1 a^2 + k_2 \ell^2)}}$$

Illustration 34. A vertical U-tube of uniform cross-section contains water upto a height of 30 cm. Show that if the water on one side is depressed and then released, its motion up and down the two sides of the tube is simple harmonic. Calculate its period.

Solution

Figure shows a U-tube of uniform cross-sectional area A . Let the liquid be depressed through the distance y in a limb, the difference of levels between two limbs will be $2y$ as shown in figure.

The liquid now oscillates about the initial positions.

Excess pressure on whole liquid

$$= (\text{excess height of the liquid column}) (\text{density}) (g)$$

$$= 2y \times 1 \times g \text{ (as density of water = 1)}$$

Restoring Force on the liquid

$$= \text{Pressure} \times \text{area of cross-section} = 2ygA$$

Due to this force the liquid accelerates and if its acceleration is a , we have $ma = -2ygA$

$$\Rightarrow (2 \times 30 \times A) a = -2ygA \Rightarrow a = -\frac{g}{30} y$$

Hence acceleration is directly proportional to displacement, so the motion is simple harmonic motion. Thus the time period T is given by

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\left(\frac{30}{g}\right)} = 2\pi \sqrt{\left(\frac{30}{980}\right)} \left[\text{as } \omega = \sqrt{\frac{g}{30}} \right] = 1.098 \text{ second}$$

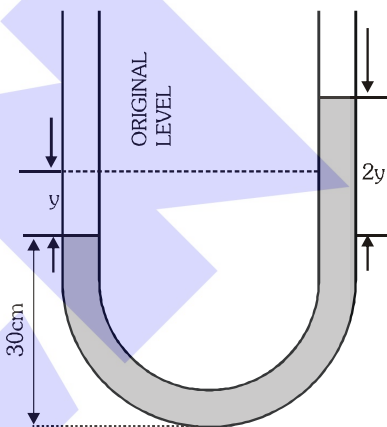
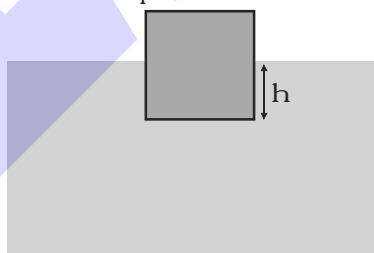
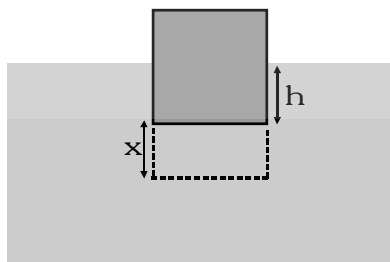


Illustration 35. Calculate the period of small oscillations of a floating box as shown in figure, which was slightly pushed down in vertical direction. The mass of box is m , area of its base is A and the density of liquid is ρ . The resistance of the liquid is assumed to be negligible.



Solution

Initially when box is floating in liquid, if its h depth is submerged in liquid then buoyancy force on it is $F_B = \text{weight of liquid displaced} = Ah\rho g$



As the box is in equilibrium, we have $Ah\rho g = mg$

Now if box is further pushed down by a distance x , net restoring force on it in upward (toward mean position) direction is

$$F_R = -[A(h+x)\rho g - mg] = -Ax\rho g \text{ [as } mg = Ah\rho g]$$

If a is the acceleration of box in upward direction we have $a = -\left(\frac{A\rho g}{m}\right)x$

Equation shows that the box executes SHM with angular frequency ω given as $\omega = \sqrt{\frac{A\rho g}{m}}$

Thus time period of its oscillation can be given as $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{A\rho g}}$

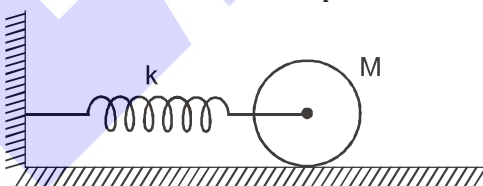
BEGINNER'S BOX-5

Super Position of SHM and Various Situations in SHM

- A particle is subjected to two simple harmonic motions along x and y directions according to $x = 3 \sin 100 \pi t$, $y = 4 \sin 100 \pi t$.
 (A) Motion of particle will be on ellipse traversing it in clockwise direction
 (B) Motion of particle will be on a straight line with slope $4/3$
 (C) Motion will be a simple harmonic motion along x -axis with amplitude 5
 (D) Phase difference between two motions is $\pi/2$
- A 20 gm particle is subjected to two simple harmonic motions
 $x_1 = 2 \sin 10 t$; $x_2 = 4 \sin\left(10t + \frac{\pi}{3}\right)$, where x_1 & x_2 are in metre & t is in sec.
 (A) The displacement of the particle at $t = 0$ will be $2\sqrt{3}$ m
 (B) Maximum speed of the particle will be $20\sqrt{7}$
 (C) Magnitude of maximum acceleration of the particle will be $200\sqrt{7}$ m/s².
 (D) Energy of the resultant motion will be 28 J

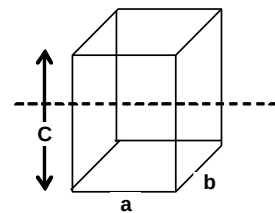
Comprehension

A solid cylinder attached with a horizontal massless spring (see figure) can roll without slipping along a horizontal surface. If the system is released from rest at a position in which spring is stretched by 0.3 m.

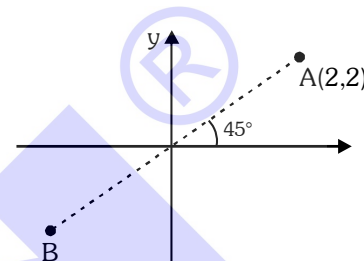


- Period of oscillation of c.o.m of cylinder is
 (A) $2\pi\sqrt{\frac{M}{k}}$ (B) $2\pi\sqrt{\frac{3M}{2k}}$ (C) $2\pi\sqrt{\frac{2M}{3k}}$ (D) $2\pi\sqrt{\frac{M}{2k}}$
- If $k = 20$ N/m, Rotational K.E. of cylinder as it passes through mean position is
 (A) 0.3 J (B) 0.45 J (C) 0.6 J (D) 0.9 J
- Translation K.E. of cylinder as it passes through mean position is
 (A) 0.3 J (B) 0.45 J (C) 0.6 J (D) 0.9 J

6. A metallic cuboid floats in equilibrium, if it is further pushed and released it starts oscillating. ρ is the density of cuboid and d is the density of liquid. Find the time period of small oscillation.



7. A particle of mass $m = 2$ kg executes SHM in xy -plane between points A and B as shown in figure under action of force $\vec{F} = F_x \hat{i} + F_y \hat{j}$. Minimum time taken by particle to move from A to B is 1 sec. At $t = 0$ the particle is at $x = 2$ and $y = 2$. Then find the value of F_x as the function of time.



12.0 DAMPED HARMONIC MOTION

SL AL

A particle will execute a simple harmonic motion with a constant amplitude if the resultant force on it is proportional to the displacement and is directed opposite to it. Nature provides a large number of situations in which such restoring force acts. The spring-mass system and the simple pendulum are examples. However, in many of the cases some kind of damping force is also present with the restoring force. The damping force may arise due to friction between the moving parts, air resistance or several other causes. The damping force is a function of speed of the moving system and is directed opposite to the velocity. Energy is lost due to the negative work done by the damping force and the system comes to a halt in due course.

The damping force may be a complicated function of speed. In several cases of practical interest the damping force is proportional to the speed. This force may then be written as

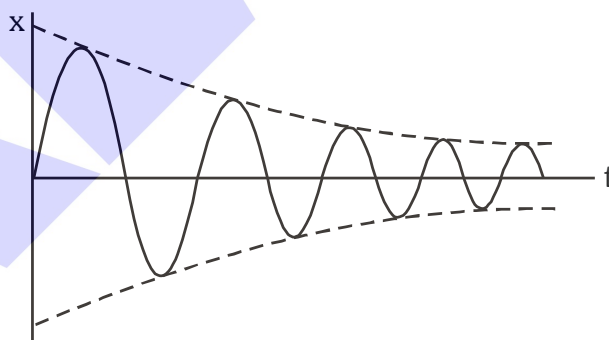
$$F = -bv.$$

The equation of motion is

$$m \frac{dv}{dt} = -kx - bv$$

This equation can be solved using standard methods of calculus.

For small damping the solution is of the form



$$x = A_0 e^{-\frac{bt}{2m}} \sin(\omega' t + \delta) \quad \dots (1)$$

$$\text{where } \omega' = \sqrt{(k/m) - (b/2m)^2} = \sqrt{\omega_0^2 - (b/2m)^2}$$

Now the mechanical energy of the undamped oscillator is $\frac{1}{2} kA^2$. For a damped oscillator, the amplitude is not constant but depends on time. For small damping, we may use the same expression but regard the amplitude as $A_0 e^{-bt/2m}$.

$$E(t) = \frac{1}{2} k A_0^2 e^{-bt/m}$$

12.1 Forced Oscillation and Resonance

SL AL

In certain situations apart from the restoring force and the damping force, there is yet another force applied on the body which itself changes periodically with time. As a simplest case suppose a force $F = F_0 \cos(\omega t)$ is applied to a body of mass m on which a restoring force $-kx$ and a damping force $-bv$ is acting. The equation of motion for such a body is

$$m \frac{dv}{dt} = -kx - bv + F_0 \cos(\omega t)$$

The motion is somewhat complicated for some time and after this the body oscillates with the frequency ω of the applied periodic force. The displacement is given by

$$x = A \cos(\omega t + \phi).$$

Such an oscillation is called forced oscillation. The amplitude of the oscillation is given by

$$A = \frac{F_0 / m}{\sqrt{(\omega^2 - \omega_0^2)^2 + (b\omega / m)^2}} \quad \dots(2)$$

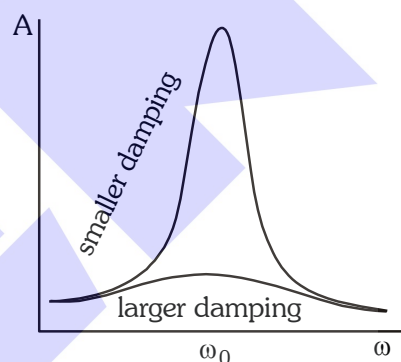
where ω_0 is the natural angular frequency.

In forced oscillation the energy lost due to the damping force is compensated by the work done by the applied force. The oscillations with constant amplitude are, therefore, sustained.

If we vary the angular frequency ω of the applied force, this amplitude changes and becomes maximum when $\omega = \omega_0$. This condition is called resonance. Resonance occurs when the applied frequency is (almost) equal to the natural frequency.

Figure shows the amplitude as a function of the applied frequency.

We see that the amplitude is large if the damping is small. Also the resonance is sharp in this case, that is the amplitude rapidly falls if ω is different from ω_0 .



If the damping were ideally zero, the amplitude of the forced vibration at resonance would be infinity by equation. Some damping is always present in mechanical systems and the amplitude remains finite.

Illustrations

Illustration 36. The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5s. In another 10s it will decrease to a times its original magnitude, where a equals

Solution

Amplitude of damped oscillator

$$A = A_0 e^{-\frac{bt}{2m}}$$

$$\text{After 5s, } 0.9 A_0 = A_0 e^{-\frac{b(5)}{2m}} \quad \dots(1)$$

After 10 more second,

$$A = A_0 e^{-\frac{b(15)}{2m}}$$

$$A = A_0 \left(e^{-\frac{b(5)}{2m}} \right)^3 \quad \dots(2)$$

From Eqs. (1) and (2),

$$A = 0.729 A_0 = \alpha A_0$$

Hence, $\alpha = 0.729$

Illustration 37. Five particles undergo damped harmonic motion. Values for the spring constant k , the damping constant b , and the mass m are given below. Which leads to the smallest rate of loss of mechanical energy at the initial moment?

- (A) $k = 100 \text{ N/m}$, $m = 50\text{g}$, $b = 8\text{g/s}$
 (C) $k = 150 \text{ N/m}$, $m = 10\text{g}$, $b = 8\text{g/s}$

- (B) $k = 150 \text{ N/m}$, $m = 50\text{g}$, $b = 5\text{g/s}$
 (D) $k = 200 \text{ N/m}$, $m = 8\text{g}$, $b = 6\text{g/s}$

Ans. (B)

Sol. $E = \frac{1}{2} k A_0^2 e^{-2\gamma t}$

$$\frac{dE}{dt} = \frac{1}{2} k A_0^2 (-2\gamma) e^{-2\gamma t}$$

$$\left| \frac{dE}{dt} \right| \propto \gamma k, \quad \gamma = \frac{b}{2m}$$

$$\left| \frac{dE}{dt} \right| \rightarrow \text{smaller}$$

Smaller will be loss.

Illustration 38. The angular frequency of the damped oscillator is given by, $\sqrt{\frac{k}{m} - \frac{r^2}{4m^2}}$ where k is the spring constant, m is the mass of the oscillator and r is damping constant. If the ratio $\frac{r^2}{mk}$ is 8%, the change in time period composed to the undamped oscillator is approximately as follows:

- (A) decreases by 8% (B) decreases by 1%
 (C) increases by 1% (D) increases by 8%

Ans. (C)

Sol. $\omega = \sqrt{\frac{k}{m} \left(1 - \frac{r^2}{4mk} \right)}$ $\frac{r^2}{mk} = \frac{8}{100}$

$$\omega = \sqrt{\frac{k}{m} \left(1 - \frac{8}{100} \right)} = \sqrt{0.98 \frac{k}{m}}$$

For undamped

$$T_1 = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$

For damped

$$T_2 = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{0.98k}}$$

CHAPTER AT A GLANCE

SIMPLE HARMONIC MOTION (S.H.M.)

In Linear S.H.M.

The resultant force (or acceleration) acting on the particle should always be proportional to the displacement of the particle and directed towards the equilibrium position

$$\therefore \vec{F} \propto -\vec{x} \text{ or } \vec{a} \propto -\vec{x}$$

In Angular S.H.M.

The resultant torque (or angular acceleration) acting on the particle should always be proportional to the angular displacement of the particle and directed towards the equilibrium position

$$\therefore \vec{\tau} \propto -\vec{\theta} \text{ or } \vec{\alpha} \propto -\vec{\theta}$$

Equation of simple Harmonic motion

In linear S.H.M.

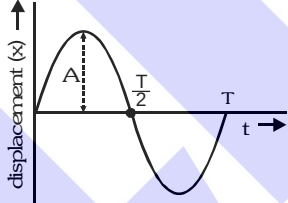
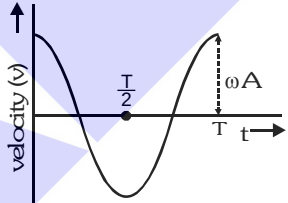
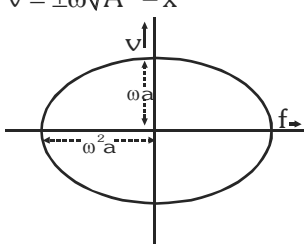
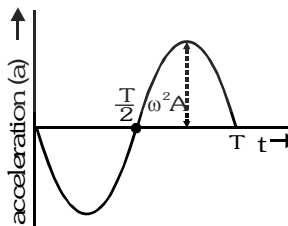
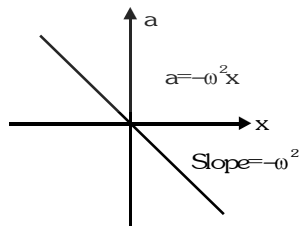
$$x = A \sin(\omega t + \phi)$$

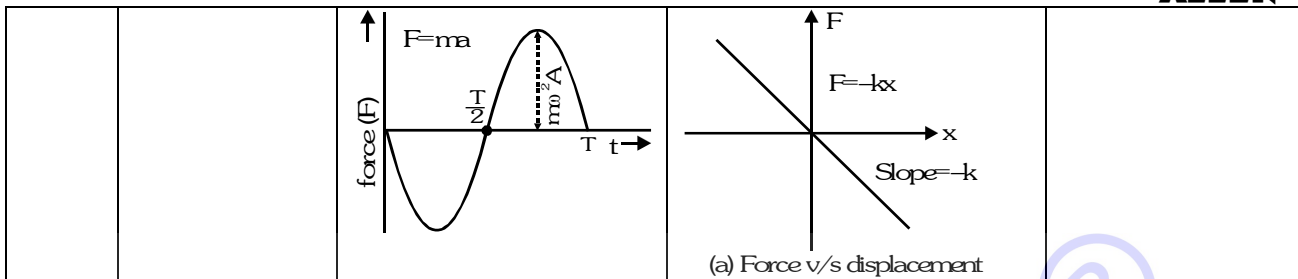
$$\{\omega^2 = k/m\}$$

In Angular SHM

Restoring torque acting on the particle where C is a constant which can be defined as torque per unit angular displacement.

$$\frac{d^2\theta}{dt^2} + \omega^2\theta = 0 \Rightarrow \omega = \sqrt{\frac{C}{I}}$$

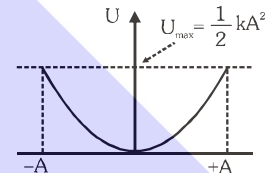
S. No.	Graph	In form of t	In form of x	Maximum value
	Displacement	$x = A \sin \omega t$ 	$x = x$	$x = \pm A$
	Velocity	$v = A\omega \cos \omega t$ 	$v = \pm \omega \sqrt{A^2 - x^2}$ 	$v = \pm \omega A$
	Acceleration	$a = -\omega^2 A \sin \omega t$ 	$a = -\omega^2 x$  (a) Acceleration v/s displacement	$a = \pm \omega^2 A$
	Force ($F = ma$)	$F = -m\omega^2 A \sin \omega t$	$F = -m\omega^2 x$	$F = \pm m\omega^2 A$

**ENERGY OF PARTICLE IN S.H.M****Potential Energy (U or P.E.)**

$$(i) \quad U = \frac{1}{2} kx^2 + U_0$$

Where the potential energy at equilibrium position = U_0

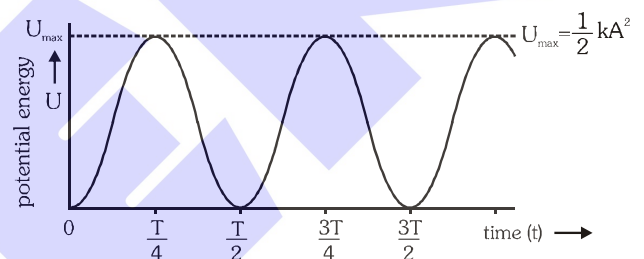
when $U_0 = 0$ then $U = \frac{1}{2} kx^2$



$$(ii) \quad x = A \sin(\omega t + \phi), \quad U = \frac{1}{2} kA^2 \sin^2(\omega t + \phi)$$

If initial phase (ϕ) is zero then

$$U = \frac{1}{2} kA^2 \sin^2 \omega t = \frac{1}{2} m\omega^2 A^2 \sin^2 \omega t$$

**Kinetic Energy (K)**

$$(i) \quad K = \frac{1}{2} mv^2 = \frac{1}{2} m\omega^2 (A^2 - x^2) = \frac{1}{2} k(A^2 - x^2)$$

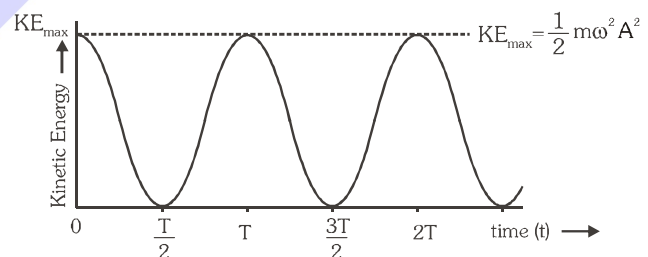
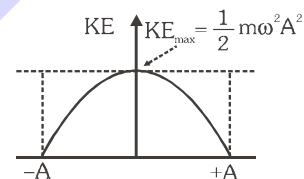
(ii) In terms of time

$$v = A\omega \cos(\omega t + \phi)$$

$$K = \frac{1}{2} m\omega^2 A^2 \cos^2(\omega t + \phi)$$

If initial phase ϕ is zero

$$K = \frac{1}{2} m\omega^2 A^2 \cos^2 \omega t$$

**Total energy (E)**

Total energy in S.H.M. is given by ; $E = \text{potential energy} + \text{kinetic energy} = U + K$

$$E = \frac{1}{2} kA^2 = \text{constant}$$

Average energy in S.H.M.

(i) The time average of P.E. and K.E. over one cycle is

$$(a) \langle KE \rangle_t = \frac{1}{4} m\omega^2 A^2 = \frac{1}{4} kA^2 \quad (b) \langle PE \rangle_t = \frac{1}{4} m\omega^2 A^2 = \frac{1}{4} kA^2$$

(ii) The position average of P.E. and K.E. between $x = -A$ to $x = A$

$$(a) \langle KE \rangle_x = \frac{1}{3} kA^2 \quad (b) \langle PE \rangle_x = U_0 + \frac{1}{6} kA^2$$

Spring Pendulum

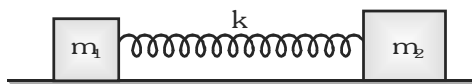
- For a spring pendulum

$$\text{Time period } T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}, \text{ Frequency, } n = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

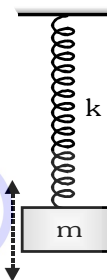
Time period of a spring pendulum is independent of Amplitude and presence of a constant force.

- If two masses m_1 and m_2 are connected by a spring and made to oscillate then time period

$$T = 2\pi\sqrt{\frac{\mu}{k}}$$



$$\text{Here, } \mu = \frac{m_1 m_2}{m_1 + m_2} = \text{reduced mass}$$



SIMPLE PENDULUM

If a heavy point mass is suspended by a weightless, inextensible and perfectly flexible string from a rigid support, then this arrangement is called a simple pendulum

Expression for time period

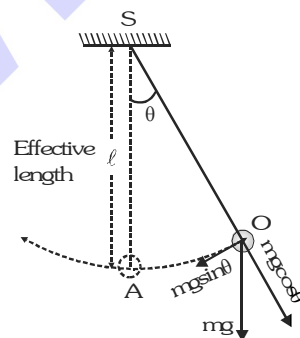
$$\omega = \sqrt{\frac{g}{\ell}} \therefore T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{\ell}{g}}$$

$$\text{If } \ell \text{ is comparable to radius of earth then time period } T = 2\pi\sqrt{\frac{R_e}{1 + \frac{R_e}{\ell}}g}$$

- The time period of oscillation of simple pendulum of infinite length ($\ell \rightarrow \infty$)

$$T = 2\pi\sqrt{\frac{R}{g}} \approx 84.6 \text{ minute} \approx 1 \frac{1}{2} \text{ hour (It is maximum time period)}$$

- If angular amplitude (θ_0) is large ($\theta_0 > 15^\circ$) then time period is given by $T = 2\pi\sqrt{\frac{\ell}{g}} \left[1 + \frac{\theta_0^2}{16} \right]$ here θ_0 is in radian.



COMPOUND PENDULUM

$$\text{Time period of compound pendulum } T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I_s}{mg\ell}}$$

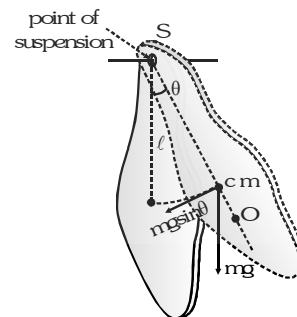
m = mass of the body

ℓ = distance between point of suspension and centre of mass

I_s = moment of inertia about horizontal axis passes through point of suspension

For minimum time period, $\ell = K$ = radius of gyration for rotation about com

$$T_{\min} = T = 2\pi\sqrt{\frac{\frac{K^2}{K} + K}{g}} = 2\pi\sqrt{\frac{2K}{g}}$$



SOME WORKED OUT ILLUSTRATIONS

Illustration 1.

A simple pendulum has time period 2s. The point of suspension is now moved upward according to relation $y = (6t - 3.75 t^2)$ m where t is in second and y is the vertical displacement in upward direction. The new time period of simple pendulum will be

- (A) 2s (B) 1s (C) 4s (D) None of these

Ans. (C)

Solution

$$T_1 = 2s = 2\pi\sqrt{\frac{\ell}{g}}, T_2 = 2\pi\sqrt{\frac{\ell}{g'}}$$

$$\text{where } g' = g + \frac{d^2y}{dt^2} = 10 - 7.5 = 2.5 = \frac{g}{4}$$

$$\Rightarrow T_2 = 2\pi\sqrt{\frac{\ell}{g/4}} = 2 \times 2 = 4s$$

Illustration 2.

Pendulum A is a physical pendulum made from a thin rigid and uniform rod whose length is ℓ . One end of this rod is attached to the ceiling by a frictionless hinge so that rod is free to swing back and forth.

Pendulum B is a simple pendulum whose length is also ℓ . The ratio $\frac{T_A}{T_B}$ for small angular oscillations-

- (A) $\sqrt{\frac{3}{2}}$ (B) $\sqrt{\frac{2}{3}}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

Ans. (B)

Solution

$$T_A = 2\pi\sqrt{\frac{I}{mg\ell}} = 2\pi\sqrt{\frac{m\ell^2/3}{mg\ell/2}} = 2\pi\sqrt{\frac{2\ell}{3g}}$$

$$\& T_B = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow \frac{T_A}{T_B} = \sqrt{\frac{2}{3}}$$

Illustration 3.

A point particle of mass 0.1 kg is executing SHM of amplitude 0.1 m. When the particle passes through the mean position, its KE is 8×10^{-3} J. Find the equation of motion of this particle if the initial phase of oscillation is 45° .

- (A) $y = 0.1 \cos(3t + \pi/4)$ (B) $y = 0.1 \sin(6t + \pi/4)$ (C) $y = 0.1 \sin(4t + \pi/4)$ (D) $y = 0.1 \cos(4t + \pi/4)$

Ans. (C)

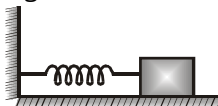
Solution

$$\text{At mean position KE} = \frac{1}{2}m\omega^2a^2 = 8 \times 10^{-3} \Rightarrow \omega = 4 \text{ rad/sec}$$

$$\text{Now let equation of SHM be } y = 0.1 \sin(4t + \phi). \text{ At } t=0, \phi = 45^\circ = \frac{\pi}{4}. \text{ Therefore } y = 0.1 \sin\left(4t + \frac{\pi}{4}\right)$$

Illustration 4.

Spring of spring constant 1200 Nm^{-1} is mounted on a smooth frictionless surface and attached to a block of mass 3 kg. Block is pulled 2 cm to the right and released. The angular frequency of oscillation is



- (A) 5 rad/sec (B) 30 rad/sec (C) 10 rad/sec (D) 20 rad/sec

Ans. (D)

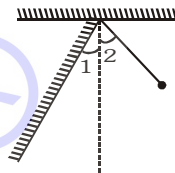
Solution

$$\text{Angular frequency } \omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{1200}{3}} = 20 \text{ rad / sec}$$

Illustration 5.

A simple pendulum of length 1m is allowed to oscillate with amplitude 2° . It collides elastically with a wall inclined at 1° to the vertical. Its time period will be : (use $g = \pi^2$)

- (A) $2/3$ sec (B) $4/3$ sec
(C) 2 sec (D) None of these



Ans. (B)

Solution

$$\text{Time period for half part : } T = 2\pi\sqrt{\frac{\ell}{g}} = 2\pi\sqrt{\frac{1}{\pi^2}} = \frac{2\pi}{\pi} = 2 \text{ sec.}$$

$$\text{So } 2^\circ \text{ part will be covered in a time } t = \frac{T}{2} = 1 \text{ sec.}$$

$$\text{For the left } 1^\circ \text{ part : } \theta = \theta_0 \sin(\omega t) \Rightarrow 1^\circ = 2^\circ \sin\left(\frac{2\pi}{T} \times t\right) \Rightarrow \frac{1}{2} = \sin\left(\frac{2\pi}{2} \times t\right) \Rightarrow \frac{\pi}{6} = \pi \times t \Rightarrow t = 1/6 \text{ sec.}$$

$$\text{Total time} = \frac{T}{2} + 2t \Rightarrow 1 + 2 \times \frac{1}{6} = 1 + \frac{1}{3} = \frac{4}{3} \text{ sec.}$$

Illustration 6.

Values of the acceleration $\ddot{x}$ of a particle moving in simple harmonic motion as a function of its displacement x are given in the table below.

$\ddot{x} (\text{mm} / \text{s}^2)$	16	8	0	-8	-16
$x (\text{mm})$	-4	-2	0	2	4

The period of the motion is

- (A) $\frac{1}{\pi}$ s (B) $\frac{2}{\pi}$ s (C) $\frac{\pi}{2}$ s (D) π s

Ans. (D)

Solution

$$\text{In SHM } a = -\omega^2 x. \text{ So } 16 = -\omega^2(-4) \Rightarrow \omega = 2 \Rightarrow \text{Time period } T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi \text{ s}$$

Illustration 7..

The time period of small oscillations of mass m :-

- (A) $2\pi\sqrt{\frac{m}{6K}}$ (B) $2\pi\sqrt{\frac{11m}{6K}}$
(C) $2\pi\sqrt{\frac{6m}{11K}}$ (D) $2\pi\sqrt{\frac{m}{K}}$

Ans. (B)

Solution

Equivalent spring constant

$$\frac{1}{K_{eq}} = \frac{1}{3K} + \frac{1}{2K} + \frac{1}{K} \Rightarrow K_{eq} = \frac{6K}{11} \therefore T = 2\pi\sqrt{\frac{m}{K_{eq}}} = 2\pi\sqrt{\frac{11m}{6K}}$$

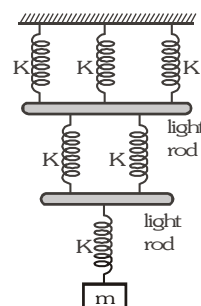


Illustration 8.

The time period of oscillation of a mass m suspended from a spring is 2 seconds. If another 2 kg mass is also suspended along with m the time period increases by 1 second. The value of mass m is

- (A) 2 kg (B) 1 kg (C) 1.6 kg (D) 2.6 kg

Ans. (C)

Solution

$$\text{Here } 2\pi\sqrt{\frac{m}{k}} = 2\text{s and } 2\pi\sqrt{\frac{m+2}{k}} = 3\text{s}$$

$$\Rightarrow \frac{3}{2} = \sqrt{\frac{m+2}{m}} \Rightarrow \frac{9}{4} = \frac{m+2}{m} \text{ which yields } m = 1.6 \text{ kg}$$

Illustration 9.

The angle made by the string of a simple pendulum with the vertical depends upon time as $\theta = \frac{\pi}{90} \sin \pi t$.

Find the length (in m) of the pendulum if $g = \pi^2 \text{ m/s}^2$

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. (A)

Solution

$$\text{Here } T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi} = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow \frac{\ell}{g} = \frac{1}{\pi^2} \text{ But } g = \pi^2 \text{ therefore } \ell = 1 \text{ m}$$

Illustration 10.

A block is in equilibrium at rest in a lift as shown in figure. Now lift starts accelerating. The minimum magnitude of acceleration of lift for which block will touch the floor of lift, is

- (A) $\frac{kh}{M}$ (B) $\frac{2kh}{M}$
(C) $\frac{kh}{2M}$ (D) $\frac{3kh}{2M}$

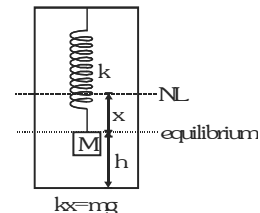
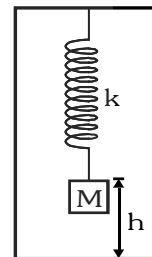
Ans. (C)

Solution

From work energy theorem $W_g + W_{\text{pseudo}} + W_{\text{spring}} = \Delta KE$

$$\Rightarrow Mgh + Mah + \frac{1}{2}kx^2 - \frac{1}{2}k(h+x)^2 = 0 \text{ where } Mg = kx$$

$$\Rightarrow Mgh + Mah - \frac{1}{2}kh^2 - khx = 0 \therefore a = \frac{kh}{2M}$$

**Illustration 11.**

Time period of a spring mass system can be changed by

- (A) changing the mass
(B) accelerating the point of suspension of the block
(C) cutting the spring (i.e. changing the length of the spring)
(D) immersing the mass in a liquid

Ans. (A,C,D)

Solution

Time period of spring mass system $T = 2\pi\sqrt{\frac{M}{K}}$ Which can be changed by (A), (C) & (D)

Illustration 12.

The time period of a particle executing SHM is T . After a time $T/6$ after it passes its mean position, its :

- (A) displacement will be one-half of its amplitude
(B) velocity will be one-half of its maximum velocity
(C) kinetic energy = $1/3$ (potential energy)
(D) acceleration will be $\frac{\sqrt{3}}{2}$ times of its maximum acceleration

Ans. (B,C,D)

Solution

Let $x = A \sin \omega t$ where $\omega = \frac{2\pi}{T}$

For (A) : At $t = \frac{T}{6}$, $x = A \sin\left(\frac{2\pi}{T}\right)\left(\frac{T}{6}\right) = A \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}A$, $x = A \sin\left(\frac{2\pi}{T}\right)\left(\frac{T}{6}\right) = A \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}A$

For (B) : $v = \frac{dx}{dt} = A\omega \cos \omega t$ At $t = \frac{T}{6}$, $v = A\omega \cos\left(\frac{2\pi}{T}\right)\left(\frac{T}{6}\right) = A\omega \cos \frac{\pi}{3} = \frac{A}{2}$

For (C) : $KE = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{v_{\max}}{2}\right)^2 = \frac{1}{4}(KE)_{\max} = \frac{1}{4}(TE)$ & $PE = TE - KE = \frac{3}{4}TE \Rightarrow KE = \frac{1}{3}(PE)$

For (D) : Acceleration $a = \frac{dv}{dt} = -A\omega^2 \sin \omega t = -\omega^2 x$

Illustration 13.

The potential energy of a particle of mass 100 g, moving along the x-axis, is given by $U = 5x(x-4)$ J, where x is in metre. Select correct alternative(s).

- (A) The particle execute SHM with mean position at $x=4$ m
- (B) The particle execute SHM with mean position at $x=2$ m
- (C) The particle execute SHM with time period $\pi/5$ second
- (D) The particle execute SHM with time period $\pi/10$ second

Ans. (B,C)

Solution

$U = 5x^2 - 20x \Rightarrow F = -\frac{dU}{dx} = -10x + 20 = -10(x-2)$

Acceleration $a = \frac{F}{m} = -100(x-2)$ so $\omega^2 = 100 \Rightarrow \omega = 10$

Time period $T = \frac{2\pi}{\omega} = \frac{2\pi}{10} = \frac{\pi}{5}$ sec

Illustration 14.

A particle moves along the Z-axis according to the equation $z = 5 + 12 \cos\left(2\pi t + \frac{\pi}{2}\right)$, where z is in cm and t is in seconds. Select the correct alternative (s)-

- (A) The motion of the particle is SHM with mean position at $z = 5$ cm
- (B) The motion of the particle is SHM with extreme position at $z = -7$ cm and $z = +17$ cm.
- (C) Amplitude of SHM is 13 cm
- (D) Amplitude of SHM is 12 cm

Ans. (A,B,D)

Solution

$z = 5 + 12 \cos\left(2\pi t + \frac{\pi}{2}\right) = 5 - 12 \sin(2\pi t) \Rightarrow (z-5) = -12 \sin 2\pi t \Rightarrow$ SHM with mean position $z = 5$ cm

Extreme position $z_{\max} = 5 + 12 = 17$ cm, $z_{\min} = 5 - 12 = -7$ cm Amplitude of SHM = 12 cm

Illustration 15.

For a SHM with given angular frequency, two arbitrary initial conditions are necessary and sufficient to determine the motion completely. These initial conditions may be

- (A) Amplitude and initial phase
- (B) Amplitude and total energy of oscillation
- (C) Initial phase and total energy of oscillation
- (D) Initial position and initial velocity

Ans. (A,C,D)

Solution

$x = A \sin(\omega t + \phi)$ As ω is given, to describe the motion completely we need A and ϕ .

Illustration 16.

The motion of a particle is represented by the function " $A \sin^2 \omega t + B \cos^2 \omega t + C \sin \omega t \cos \omega t$ " where x , ω and t represent displacement, angular frequency and time respectively. The constants A , B and C are related to the amplitudes of displacement. Select **CORRECT** alternative(s).

- (A) The motion of particle is SHM when $A=B$.
 (B) The motion of particle is SHM when $A=B$ and $C=0$.
 (C) If $B = \frac{C}{2} = -A$, then the amplitude of SHM is $B\sqrt{2}$.
 (D) If $A=B = \frac{C}{2}$, then the axis of vibration of SHM shifts by a distance B towards $+x$ axis.

Ans. (ACD)**Solution**

If $A=B$ then $x = A + C \sin \omega t \cos \omega t = A + \frac{C}{2} \sin 2\omega t \Rightarrow$ SHM

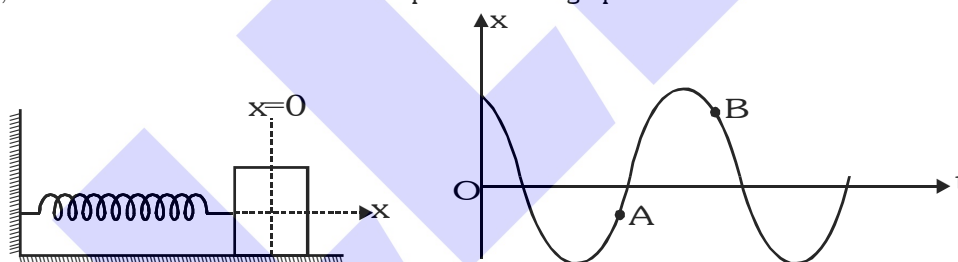
If $A = B$ & $C = 0$ then $x = A \Rightarrow$ along a straight line.

If $B = \frac{C}{2} = -A$ then $x = B \cos 2\omega t + B \sin 2\omega t \Rightarrow$ amplitude $= B\sqrt{2}$

If $A = B = \frac{C}{2}$ then $x = B + B \sin 2\omega t \Rightarrow$ Axis of vibration of SHM shifts by a distance B towards $+x$ -axis.

Illustration 17 to 19.

A 100 g block is connected to a horizontal massless spring of force constant 25.6 N/m. The block is free to oscillate on a horizontal frictionless surface. The block is displaced 3 cm from the equilibrium position, and at $t=0$, it is released from rest at $x=0$. The position-time graph of motion of the block is shown in figure.



17. When the block is at position A on the graph, its
 (A) position and velocity both are negative. (B) position and velocity both are positive.
 (C) position is negative and velocity is positive. (D) position is positive and velocity is negative.
18. When the block is at position B on the graph, its
 (A) position and velocity both are negative. (B) position and velocity both are positive.
 (C) position is negative and velocity is positive. (D) position is positive and velocity is negative.
19. Let us now make a slight change to the initial conditions. At $t=0$, let the block be released from the same position with an initial velocity $v_i = 64$ cm/s. Position of the block as a function of time can be expressed as
 (A) $x = 5 \sin 16t$ (B) $x = 5 \sin(16t + 37^\circ)$ (C) $x = 5 \sin(16t - 37^\circ)$ (D) $x = 5 \cos(16t + 37^\circ)$

Solution**17. Ans. (C)**

At position A, x is negative and $\frac{dx}{dt}$ is positive.

18. Ans. (D)

At position B, x is positive and $\frac{dx}{dt}$ is negative.

19. Ans. (B)

Let $x = A \sin(16t + \phi)$ $v = A\omega \cos(16t + \phi)$ where $\omega^2 = \frac{K}{m} = \frac{25.6}{0.1} = 256 \Rightarrow \omega = 16 \text{ rad/sec}$

At $t=0$, $3 = A \sin \phi$ & $64 = 16A \cos \phi \Rightarrow \tan \phi = \frac{3}{4} \Rightarrow \phi = 37^\circ$ Also $A = 5 \text{ cm}$

Therefore equation of SHM $x = 5 \sin(16t + 37^\circ)$

Illustration 20 to 22.

X_1 and X_2 are two points on the path of a particle executing SHM in a straight line, at which its velocity is zero. Starting from a certain point X ($X_1X < X_2X$) the particle crosses this point again at successive intervals of 2s and 4s with a speed of 5 m/s.

20. The time period of SHM is

- (A) 6 s (B) 3s (C) 4s (D) 2s

21. The amplitude of SHM is

- (A) $10\sqrt{3} \text{ m}$ (B) $\frac{10}{\sqrt{3}\pi} \text{ m}$ (C) $\frac{10\sqrt{3}}{\pi} \text{ m}$ (D) can't be determined

22. The maximum speed of the particle is

- (A) $\frac{10\sqrt{3}}{\pi} \text{ m/s}$ (B) $10\sqrt{3} \text{ m/s}$ (C) $\frac{\sqrt{3}}{10} \text{ m/s}$ (D) $\frac{10}{\sqrt{3}} \text{ m/s}$

Solution

20. Ans. (A)



21. Ans. (C)

The given SHM can be represented in circular motion

In shown diagram $\theta = \left(\frac{2\pi}{6}\right)(2) = \frac{2\pi}{3}$

Let $OX = x_0$ and amplitude $= OP = A$

Now $\frac{x_0}{A} = \cos \frac{\theta}{2}$ so $x_0 = \frac{A}{2}$

As $v = \omega \sqrt{A^2 - x_0^2}$ so $5 = \frac{2\pi}{6} \sqrt{A^2 - \frac{A^2}{4}} \Rightarrow A = \frac{10\sqrt{3}}{\pi} \text{ m}$

22. Ans. (D)

Maximum speed $= A\omega = \left(\frac{10\sqrt{3}}{\pi}\right)\left(\frac{2\pi}{6}\right) = \frac{10}{\sqrt{3}} \text{ m/s}$

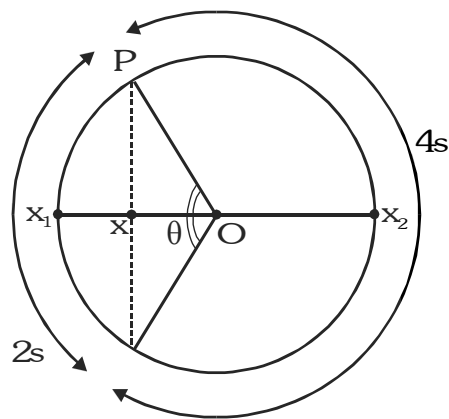
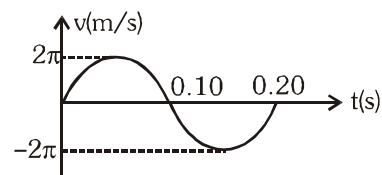


Illustration 23.

A simple harmonic oscillator consists of a block attached to a spring with $k = 200 \text{ N/m}$. The block slides on a frictionless horizontal surface, with equilibrium point $x = 0$. A graph of the block's velocity v as a function of time t is shown (use $\pi^2 = 10$)



Column I

- (A) The block's mass in kg
 (B) The block's displacement at $t = 0$ in meters
 (C) The block's acceleration at $t = 0.10$ s in m/s^2
 (D) The block's maximum kinetic energy in joule

Column II

- (P) -0.20
 (Q) -200
 (R) $+0.20$
 (S) $+4.0$
 (T) $+200$

Ans. (A) R, (B) P, (C) Q, (D) S

Solution

$$v_m = A\omega \Rightarrow A = \frac{v_m}{\omega} = \frac{2\pi}{2\pi} \times (0.2) = 0.20 \text{ m}; T = 2\pi\sqrt{\frac{m}{k}} \Rightarrow m = \frac{T^2 k}{4\pi^2} = 0.2 \text{ kg}$$

$$\text{At } t = 0.1, \text{ acceleration is maximum} \Rightarrow -\omega^2 A = -200 \text{ m/s}^2$$

$$\text{Maximum energy} = \frac{1}{2}mv_m^2 = 4\text{J or } \frac{1}{2}kA^2 = E_{\max} = \frac{1}{2} \times 200 \times 0.04 = 4 \text{ J}$$

ANSWERS**BEGINNER'S BOX-1**

1. (C) 2. (A) 3. (D) 4. (C) 5. (C) 6. (C) 7. (C)

BEGINNER'S BOX-2

1. (A) 2. (B,C) 3. (A) 4. (D) 5. (C)
 6. (a) $T/12$ (b) $T/8$ (c) $T/6$ (d) $T/4$, (e) $T/8$ 7. $x = (10 \text{ cm}) \sin \left[\left(\frac{\pi}{3} \text{ s}^{-1} \right) t + \frac{\pi}{6} \right]$, $\frac{10}{9} \pi^2 \approx 11 \text{ cm/s}^2$
 8. (3)

BEGINNER'S BOX-3

1. (C) 2. (D) 3. (A) 4. (A) 5. (D)
 6. $\frac{5}{2\pi} \text{ Hz}$, 5 cm 7. 0.16 kg

8. (a) $2\pi\sqrt{\frac{m}{k_1+k_2}}$, $k_{\text{eq.}} = k_1 + k_2$; (b) $2\pi\sqrt{\frac{m}{k_1+k_2}}$, $k_{\text{eq.}} = k_1 + k_2$; (c) $2\pi\sqrt{\frac{m(k_1+k_2)}{k_1k_2}}$, $k_{\text{eq.}} = \frac{k_1k_2}{k_1+k_2}$

BEGINNER'S BOX-4

1. (C) 2. (A,B,D) 3. (B) 4. (A) 5. $2\pi\sqrt{\frac{r\sqrt{2}}{g}}$, $r/\sqrt{2}$

BEGINNER'S BOX-5

1. (B) 2. (A,B,C,D) 3. (B) 4. (A) 5. (C) 6. $T = 2\pi\sqrt{\frac{cp}{gd}}$
 7. $-4\pi^2 \cos \pi t$